Multi-Product Pricing under Rational Inattention: Evidence, Estimation, and Field Deployment from Airline Ancillaries

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Abstract. We study the joint pricing of close substitutes when customer attention is unevenly distributed across products and data on any single (product, context) cell are too thin for fully separate demand estimation. Airline ancillaries, product-line variants, subscription tiers, and hotel room categories all share this structure. The standard multinomial-logit benchmark imposes homogeneous substitution and therefore cannot represent the product-level asymmetries observed in such markets. We develop a generalized rational-inattention demand model with product-specific attention weights. The seller's revenue problem is convex in the induced purchase probabilities and admits a semi-closed-form solution: each product's optimal price decomposes into its effective cost, a product-specific attention markup, and a common demand term that depends on the segment only through a single scalar, the aggregate purchase probability. Pricing therefore reduces to a one-dimensional root search regardless of assortment size. Calibration reduces to a linear program over the KKT residuals of the consumer's problem, with a finite-sample error bound governed by the ℓ_1-minimum gain of the design matrix. When per-segment data are sparse, we pool neighboring segments and replace segment-specific intercepts with a linear function of observable features. The total parameter error decomposes into statistical error, merging bias, and approximation bias; the corresponding bias–variance trade-off is operationalized in deployment by a split-sample cross-validation procedure for configuration selection. A revenue-loss bound quadratic in the ℓ_1 parameter error links sample size to revenue and motivates three pricing rules of increasing data appetite: a segment-based rule, a feature-based rule, and a propensity-based rule (deployed as the PCP arm). We evaluate the framework in collaboration with a low-cost airline. Across five checked-baggage tiers, the cheapest weight tier (20 kg) is nearly two orders of magnitude ($\approx 80\times$) more price-sensitive than the highest (40 kg), and cross-tier substitution is small relative to own-price response; both patterns are inconsistent with homogeneous substitution. In a randomized field deployment over 48 origin–destination pairs, the segment-based and feature-based rules each raise baggage revenue by 7–8% ($p < 0.01$) relative to uniform pricing, and their posted-price distribution first-order stochastically dominates that of the carrier's incumbent rule: average posted prices fall while revenue rises.

Public version: partner-identifying details are omitted, and monetary results are reported as percentages of the control-arm mean.

Key words: Multi-product pricing, rational inattention, ancillary pricing, demand estimation, heterogeneous substitution, field experiments

1. Introduction

A common pricing problem runs through many revenue-management settings: a firm must jointly price several close substitutes; each customer's willingness to pay shifts with an observable purchase context such as occasion, party size, advance purchase, or time to consumption; and within any single product-by-context cell the available data are too thin to estimate demand separately. Retail product lines, software

subscription tiers, hotel room categories, insurance menus, and airline ancillaries all share this structure. The multinomial-logit (MNL) benchmark is easy to estimate and deploy, but it rules out by assumption the product-level substitution asymmetries that characterize these markets, in which customers attend more carefully to some products than to others. Addressing this setting takes three ingredients in combination: a demand model that admits product-level substitution heterogeneity, an estimation routine that remains stable under data sparsity, and a pricing rule that operations teams can interpret and deploy.

We study this problem through airline ancillary pricing, with checked baggage as the running example. Ancillary services account for a substantial share of airline revenue but are typically priced through static rule tables only loosely tied to demand (IdeaWorksCompany 2024). Checked baggage fits the three ingredients above: the carrier jointly prices several close substitutes (weight tiers); demand varies with itinerary, advance-purchase window, party size, and travel purpose; and route–segment-level data are thin. The setting offers two further advantages. First, the same nominal allowance carries different value and handling cost across itineraries, so heterogeneity is economically meaningful rather than merely statistical. Second, the carrier randomizes posted prices at the booking-session level, providing exogenous variation for both estimation and field validation. The framework developed below is not specific to airlines: the model, estimation procedure, and pricing rules apply to other multi-product settings with heterogeneous substitution and sparse context-level data.

Two further considerations shape the modeling approach. First, ancillary choices are made within a complex booking process in which passengers simultaneously evaluate itinerary, airfare, seats, and additional add-ons, so the assumption that every option receives equal attention is implausible at the outset. Rational inattention (RI) supplies the corresponding decision-theoretic primitive: customers face information-processing costs and allocate attention across products accordingly. Second, RI is well matched to the data we observe. Standard random-utility models build choice from individual-level utility maximization with i.i.d. taste shocks, whereas the observable primitives in ancillary pricing are aggregate session- or segment-level shares; RI works directly with these aggregate probabilities and encodes product-specific substitution through information-cost parameters rather than through a latent taste distribution. Both features, a behavioral primitive consistent with multi-task booking and a structure that operates directly on aggregate shares, are exploited in the pricing characterization and the finite-sample estimation bounds developed in the sequel. The carrier's randomized pricing data corroborate this modeling choice. The reduced-form analysis in §2 documents three patterns inconsistent with homogeneous substitution: own-price response is markedly stronger for lower tiers, cross-tier substitution is small relative to own-price response, and price sensitivity itself varies with booking context. These patterns motivate the generalized-RI specification developed in §3.

1.1. Contributions

The paper develops a unified framework for multi-product pricing under rational inattention, combining a structural demand model, a sparse-data estimation methodology, three implementable pricing rules, and a randomized field validation. The main contributions are as follows.

A generalized RI demand model and closed-form pricing. The demand model captures substitution heterogeneity through product-specific attention weights (§3). The induced revenue-maximization problem is convex, and the optimum admits a semi-closed-form representation (Proposition 1 and Theorem 1): each product's optimal price decomposes into its effective cost, a product-specific attention markup, and a common demand term that depends on the segment only through a single scalar, the aggregate purchase probability. Pricing therefore reduces to a one-dimensional root search per segment, independent of the assortment size.

Calibration under data sparsity. The KKT conditions of the consumer's attention problem are linear in the structural parameters once shares are observed; calibration is therefore a linear program in the KKT residuals (Theorem 2). When per-segment data are thin, the calibration is augmented by a two-level pooling scheme that borrows strength across segments through observable booking features. The resulting parameter error admits a three-term decomposition into merging bias, approximation bias, and statistical error (Theorems 3 and 4); this decomposition is operationalized through a cross-validated configuration-selection procedure in §EC.2 of the e-companion.

Three pricing rules. Three rules span the complexity–data trade-off: a segment-based rule for data-rich markets, a feature-based rule that pools segments through a feature-linear utility shifter (§5.2), and a propensity-based rule that decouples demand estimation from price selection via a one-dimensional threshold adjustment, deployed as the PCP arm in the field test (§5.3). A revenue-loss bound quadratic in the ℓ_1 parameter error (Theorem 5) links calibration error to revenue loss and orders the three rules along the complexity dimension.

Field validation. A randomized field experiment with a major low-cost carrier covers 48 origin–destination pairs over August–September 2025, evaluating the framework end to end. The segment-based and feature-based rules each raise baggage revenue by 7–8% relative to a uniform-pricing control ($p < 0.01$), and their posted-price distribution first-order stochastically dominates that of the carrier's incumbent propensity-based rule. The broader lesson is that *attention-aware personalization raises revenue by lowering the average price, not raising it*, because the gains come from realigning markups with where customers attend to price rather than from extracting more per customer.

Scope of applicability. The framework applies to settings combining four ingredients: horizontally differentiated substitute products with cross-substitution; product-level heterogeneity in price sensitivity or attention; customer-context heterogeneity observable through features; and sparse data at the (product,

context) cell level. Beyond airline ancillaries, retail product-line variants, subscription tiers, hotel room categories, and insurance menus share this profile. Settings with binding capacity constraints, intertemporal dynamics with strategic customer waiting, or product hierarchies without meaningful cross-substitution lie outside the framework as developed here.

1.2. Related Literature

The paper relates to several streams. The first is rational inattention and limited-attention discrete choice. Rational inattention originates in macroeconomics (Sims 2003) and was brought into discrete choice by Matejka and McKay (2015), who show that Shannon information costs generate multinomial-logit choice probabilities. Fosgerau et al. (2020) establish a broader equivalence between RI and entropy-perturbed discrete-choice models. Within operations management, Boyacı and Akçay (2018) study pricing when customers have limited attention, Huettner et al. (2019) characterize choice when alternatives carry different information costs, and Canyakmaz and Boyacı (2023) analyze queueing systems with rationally inattentive customers. Recent empirical and quantitative work documents that attention frictions matter in large-assortment environments (Natan 2025, Turlo et al. 2025). Our paper builds on this stream by using a *generalized* RI specification with product-specific attention weights (which sits in the Uniformly Posterior-Separable (UPS) class of Caplin et al. (2022) and reduces to the Shannon-RI logit model of Matejka and McKay (2015) when the weights are equal) as the organizing structure of a multi-product pricing problem with sparse data, explicit price formulas, and field deployment.

A second strand studies demand models with heterogeneous substitution and their pricing implications. The MNL model (McFadden 1973, Train 2009) remains the standard benchmark, while nested-logit and GEV models relax IIA through structured correlation (McFadden 1978). Alternatives include the exponomial model (Alptekinoglu and Semple 2016), mixed logit (Train 2009), latent-class specifications (Greene and Hensher 2003), and the broader perturbed-utility framework (Anderson et al. 1988, Fosgerau et al. 2013, 2020), within which Yan et al. (2022) study data-driven identification of the representative consumer. Our demand specification is the additive perturbed-utility model with a weighted-Shannon perturbation; it inherits the Bregman/UPS reading of Fosgerau et al. (2020) and Caplin et al. (2022), and it is equivalent to the Marginal Distribution Model of Mishra et al. (2014) with product-specific exponential noise marginals and a comonotone copula, as we develop in §EC.3. Relative to those papers, and to the data-driven perturbed-utility pricing analyses of Yan et al. (2022) and Liu et al. (2025), the paper's contribution lies in the combination of (i) a sparse-data LP-based calibration pipeline with an explicit three-term parameter-error decomposition, and (ii) a transparent semi-closed-form pricing ladder that scales to thousands of segments and supports a randomized field deployment, rather than in a new demand-model class.

The paper also connects to multi-product pricing under discrete choice, a central topic in revenue management. Broder and Rusmevichientong (2012) study dynamic pricing under general parametric choice models;

Li and Huh (2011) establish concavity and structural properties under MNL and nested-logit demand; Gallego and Wang (2014) analyze multi-product competition under nested logit; and Zhang et al. (2018) derive a Lambert-W characterization under GEV demand. Richer specifications such as mixed logit add flexibility at a substantial computational cost in both estimation and pricing (Li et al. 2019, Train 2009). The RI formulation we adopt sits between these poles: it moves beyond homogeneous substitution while retaining a convex pricing problem, a one-dimensional representation of the optimum, and comparative statics transparent enough for deployment.

A fourth strand studies data-driven pricing under customer heterogeneity, contextual features, and data sparsity. Cohen et al. (2020) develop feature-based dynamic pricing, Keskin et al. (2024) analyze clustering and feature-based pricing with statistical guarantees, and Tang et al. (2025) study offline feature-based pricing under censored demand. The broader literature in revenue management emphasizes the trade-off among model flexibility, statistical reliability, and implementability when prices depend on customer context (Talluri and Van Ryzin 2004, Phillips 2021). In the RI setting, this trade-off admits a decomposition into merging bias, approximation bias, and estimation error that translates directly into revenue-loss bounds.

Our estimation and pricing pipeline also connects to an OR/MS literature on pooling, clustering, and transfer learning for data-scarce decision problems. Cohen et al. (2022) study data aggregation through clustering for demand prediction and characterize the resulting flexibility–aggregation trade-off. Bernstein et al. (2019) show how dynamic clustering can pool information across customers in data-driven assortment personalization. In pricing, Keskin et al. (2024) develop a joint clustering and feature-based pricing approach for retail electricity markets, and Lei et al. (2024) study transfer-learning-based pooling for retail demand prediction. We embed pooling and feature-based approximation directly into a multi-product ancillary-pricing problem with substitutable products and link the resulting learning error to pricing performance through explicit revenue-loss bounds.

Finally, the paper relates to ancillary pricing and airline applications. Prior work has studied joint pricing of primary products and ancillaries, passenger-choice models with ancillary services, and live deployment of ancillary-pricing systems (Ødegaard and Wilson 2016, Bockelie and Belobaba 2017, Kolbeinsson et al. 2022). In adjacent airline-pricing work, Cohen et al. (2023) use a field experiment to redesign airfare ladders under competition, and Liu et al. (2025) develop pricing analytics for ancillary products using conversion data within a perturbed-utility framework. Checked baggage is a natural setting for extending this line of work: the products are horizontally differentiated, frequently repriced, and sold under sparse data at the route–segment level.

2. Motivating Evidence from Randomized Baggage Pricing

Airline checked baggage offers a useful testbed for the multi-product pricing problems described in §1: the menu is small and well defined, prices are reset frequently, and the partner carrier's booking platform

randomizes posted prices, generating the exogenous variation needed to identify substitution patterns. The evidence speaks directly to baggage demand and, more broadly, to the heterogeneous substitution patterns that any multi-product pricing system must accommodate.

2.1. Institutional Background

We collaborate with a low-cost airline (hereafter "the Airline"), whose pricing system randomly assigns posted prices to booking sessions, enabling controlled experimentation. Several checked baggage tiers are offered; on each flight leg a customer selects at most one unit of a single tier or opts out, so demand is horizontal substitution among I products plus a no-purchase option. Booking attributes (passenger count, trip type and duration, advance purchase, season, day of week, fare class) interact to generate $\sim 10^4$ origin–destination (OD) segments, which makes fully disaggregated estimation difficult.

Segment-based pricing is economically motivated: baggage is tied to the itinerary, so willingness to pay and effective service cost depend on the main product and booking state. The Airline's incumbent rule applies nearly identical additive surcharges per feature and preserves roughly constant price gaps across tiers; it is easy to implement but overly rigid. The randomized evidence below shows this benchmark is too restrictive and motivates a model with product-level heterogeneity.

2.2. Reduced-Form Evidence against Homogeneous Substitution

The Airline's pricing system assigns posted prices at the session level, independently across sessions conditional on OD, date, and observed booking features. This exogenous variation is used to test three reduced-form implications of a homogeneous-substitution benchmark; the evidence supports an RI-motivated specification but does not constitute literal identification of attention costs. The complete regression tables are reported in §EC.4 of the e-companion; compact summary statistics are reported here.

2.2.1. H1: Tier Gradient in Own-Price Sensitivity Rational inattention suggests that consumers compare most carefully when scrutiny yields the greatest benefit. In baggage pricing, the cheapest tier attracts the largest pool of potential buyers and entails the lowest switching cost, making it the natural focal point for price comparison. We therefore test whether own-price sensitivity declines monotonically across tiers. Panel A of Table 1 reports regressions of tier-specific purchase quantity on own price, controlling for covariates and fixed effects. The 20 kg tier exhibits by far the largest own-price sensitivity (-0.0401). The magnitude falls sharply for 25 kg (-0.0028) and 30 kg (-0.0017), and becomes nearly zero for 35 kg (-0.0001, insignificant). The 40 kg coefficient remains small but significant (-0.0005, $p < 0.01$), suggesting residual attention to this tier, possibly because it is salient as the highest option and a close substitute for purchasing two 20 kg units. Overall, the cheapest tier is about 14 times more price-sensitive than the 25 kg tier and about 80 times more sensitive than the 40 kg tier, while the 35 kg coefficient is statistically indistinguishable from zero.

Table 1 Reduced-form evidence on substitution across baggage tiers

	(1) Quantity_{20}	(2) Quantity_{25}	(3) Quantity_{30}	(4) Quantity_{35}	(5) Quantity_{40}
Panel A. Own price sensitivity					
Own price	−0.0401***	−0.0028***	−0.0017***	−0.0001	−0.0005***
	(0.0011)	(0.0007)	(0.0004)	(0.0002)	(0.0002)
Panel B. State-dependent own price sensitivity					
Own price	−0.0547***	−0.0046***	−0.0025***	−0.0002	−0.0008***
	(0.0013)	(0.0008)	(0.0005)	(0.0002)	(0.0002)
Own price ×	0.0098***	0.0012***	0.0005***	0.0001**	0.0002***
Number of passengers	(0.0003)	(0.0001)	(0.0001)	(0.0000)	(0.0000)
Own price ×	−0.0009***	−0.0002***	−0.0001	−0.0000	−0.0000
Months to departure	(0.0002)	(0.0001)	(0.0001)	(0.0000)	(0.0000)
Panel C. Cross price sensitivity					
Price_{20}	−0.0886***	0.0074***	0.0029***	0.0007***	0.0020***
	(0.0018)	(0.0006)	(0.0004)	(0.0002)	(0.0004)
Price_{25}	0.0819***	−0.0106***	−0.0011*	−0.0005	−0.0011**
	(0.0027)	(0.0010)	(0.0006)	(0.0003)	(0.0005)
Price_{30}	−0.0016	0.0010	−0.0034***	0.0007**	0.0001
	(0.0022)	(0.0008)	(0.0006)	(0.0003)	(0.0004)
Price_{35}	−0.0077***	0.0010	0.0014***	−0.0009***	0.0006*
	(0.0019)	(0.0008)	(0.0005)	(0.0002)	(0.0003)
Price_{40}	0.0038***	0.0003	0.0002	0.0002*	−0.0008***
	(0.0010)	(0.0004)	(0.0003)	(0.0001)	(0.0002)

Notes: This table presents the estimated price sensitivity of quantity purchased of each baggage tier. Controls include number of passengers (except Panel B), trip days, months to departure (except Panel B), airfare, as well as fixed effects of origin-destination pair, booking date, pricing features, fare class, and payment currency. Cluster-adjusted standard errors at the passenger name record (PNR) level are shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

2.2.2. H2: State-Dependent Responses Endogenous attention depends on booking context. In low-attention states such as larger group size or late booking, passengers should exhibit lower price sensitivity. In Panel B of Table 1, we interact own price with number of passengers and months to departure to allow for heterogeneous responses across different booking scenarios. We find that all interaction terms involving the number of passengers are positive and significant, indicating that own price sensitivity is decreasing (in absolute value) in the number of passengers, which confirms that larger groups are less price-sensitive. Similarly, the interaction terms involving months to departure are all negative (although only the 20 kg and 25 kg tiers are significant), suggesting that travelers closer to departure are less price-sensitive.

2.2.3. H3: Limited Cross-Tier Substitution When making the purchase decision of a certain tier, rational inattention predicts that cross-tier price sensitivity should be small relative to own price sensitivity. Customers consider a restricted set of alternatives, leading to weak substitution across baggage tiers. Panel C of Table 1 reports the estimated own and cross price sensitivity. We find that all cross price sensitivity are smaller than own price sensitivity estimates, except for the substitution from the 20 kg tier to the 40 kg tier, which is consistent with the idea that the 40 kg tier is a close substitute for two units of the 20 kg tier.

2.3. Modeling Implications

The three patterns above have direct implications for model selection. First, the steep tier gradient (Panel A) requires product-level substitution heterogeneity; under MNL's IIA restriction, all products share a common substitution parameter, and the near-two-order-of-magnitude gap between the cheapest and highest tiers cannot be reproduced. Second, the limited cross-price substitution (Panel C) identifies own-price response as the dominant economic channel, favoring product-specific own-price curvature over nesting-based tier groupings. Third, the state-dependent variation (Panel B) shows that the substitution parameters themselves vary with observable booking features, calling for product-level heterogeneity embedded within a feature-based segmentation. The next section introduces a model with these properties.

3. A Heterogeneous-Attention Pricing Model

The evidence in §2 calls for a demand system with asymmetric product responses, where the cheapest product is more than an order of magnitude more price-sensitive than the others. The pricing model developed below operationalizes that asymmetry; baggage remains the running example, but the formulation applies to a generic set of substitute products and is not specific to airlines, baggage, or ancillaries. Capturing the asymmetry requires product-specific information-cost scalars; hence, the Shannon RI model of Sims (2003) and Matejka and McKay (2015), which collapses to multinomial logit under a common scalar, serves as a nested benchmark rather than the main specification.

We instead use an additive perturbed-utility (APU) specification (Anderson et al. 1988, Fudenberg et al. 2015) with a weighted Shannon-entropy perturbation and product-specific weights. This specification is an instance of the Marginal Distribution Model with exponential noise marginals (Mishra et al. 2014) and admits a generalized rational-inattention interpretation within the Uniformly Posterior-Separable class of Caplin et al. (2022), through the Bregman information cost of Fosgerau et al. (2020). The APU form is analytically and computationally convenient, while the generalized-RI interpretation gives the weights their economic meaning as attention primitives. Section EC.3 develops both representations as complementary interpretive lenses on the same heterogeneous-α choice probabilities and locates the demand specification and structural concavity result within the perturbed-utility/MDM literature.

3.1. Generalized Rational-Inattention Demand

Let products be indexed by $i \in [I]$, with posted prices $\boldsymbol{p} = (p_1, \ldots, p_I)$ and effective marginal costs $\boldsymbol{c} = (c_1, \ldots, c_I)$. In the baggage application, these effective costs may vary across itinerary groups because handling requirements differ with the underlying flight. The no-purchase option is indexed by 0 and normalized to utility zero. Customers are partitioned into booking segments indexed by k, where each segment corresponds to an observable feature profile $\boldsymbol{x}_k$. For segment k, the deterministic utility of baggage tier i is

$$u_{ik} := v_{ik} - \gamma_k p_i,$$

where v_{ik} is the segment-specific valuation and $\gamma_k > 0$ is the common within-segment price sensitivity.

A customer in segment k chooses how much information to process before selecting among baggage tiers, paying an attention cost that is weighted separately across tiers. The induced aggregate choice shares solve the convex programming problem

$$\max_{\boldsymbol{s}_k \in \Omega} \quad \sum_{i \in [I]} s_{ik}(v_{ik} - \gamma_k p_i) - \sum_{i \in \{0\} \cup [I]} \alpha_{ik} s_{ik} \ln s_{ik}, \tag{1}$$

where $\boldsymbol{s}_k = (s_{0k}, s_{1k}, \ldots, s_{Ik})$ is the market-share vector whose entries are the purchase probabilities of the outside option and the I baggage tiers and satisfy $s_{ik} \geq 0$ with $\sum_{i=0}^{I} s_{ik} = 1$, $\alpha_{ik} > 0$ is the per-tier attention weight for option i, and $\Omega := \{A\boldsymbol{s}^\top \leq \boldsymbol{b}\}$ captures the product structure, with $A \in \mathbb{R}^{m \times (I+1)}$ having rows A_q ($q \in [m]$) and entries A_{qi} ($q \in [m]$, $i \in \{0\} \cup [I]$), and right-hand side $\boldsymbol{b} = (b_1, \ldots, b_m)^\top$. In the horizontal case studied for baggage pricing, Ω reduces to the simplex $\{\boldsymbol{s} \geq 0 : \sum_{i=0}^{I} s_i = 1\}$, so $m = 1$ and the products are pure substitutes together with the outside option. The transition from the inequality form $\{A\boldsymbol{s}^\top \leq \boldsymbol{b}\}$ to the simplex equality $\sum_i s_i = 1$ reflects the fact that the unique constraint $\sum_i s_i \leq 1$ is active at any interior optimum of (1), as established in Lemma EC.1.

The weighted-entropy term is the model's attention primitive. Smaller α_{ik} means that the customer can evaluate option i more precisely at lower informational cost, which translates into greater responsiveness to the utility of that option and, in particular, to its price. Product-specific α_{ik} values allow lower baggage tiers to be scrutinized more intensely than higher tiers, matching the empirical patterns in §2. The textbook Shannon RI model of Sims (2003) and Matejka and McKay (2015) corresponds to the limiting case $\alpha_{ik} \equiv \lambda_k$, in which (1) collapses to multinomial logit; product-specific weights are what allow the tier asymmetry documented in §2 to be expressed.

The α_{ik} are nonetheless legitimate rational-inattention primitives. Specification (1) coincides with the conditional choice probabilities of a generalized rational-inattention problem in which the information cost is the *Bregman information* (Banerjee et al. 2005, Fosgerau et al. 2020) generated by the weighted Shannon entropy $\Phi(\boldsymbol{s}) = \sum_i \alpha_{ik} s_{ik} \ln s_{ik}$ (we denote the entropy by Φ to avoid notation clash with the feasible set Ω in (1)). This places the model in the Uniformly Posterior-Separable class of Caplin et al. (2022), the axiomatically grounded generalization of Shannon RI: each α_{ik} is the per-tier weight on the agent's mutual-information bill, and Shannon RI is recovered when those weights are equal. The formal bridge is developed in §EC.3. Equivalently, (1) is the Marginal Distribution Model of Mishra et al. (2014) with product-specific exponential noise marginals and a comonotone copula, which independently licenses the convex-pricing structure derived in §3.4.

Equivalent convex formulation. Since the weighted Shannon entropy is admissible (Hofbauer and Sandholm 2002), problem (1) can also be written as the additive perturbed utility (APU) problem of Fudenberg et al. (2015) and Anderson et al. (1988),

$$\max_{\boldsymbol{s}_k \in \Omega} \quad \boldsymbol{s}_k \cdot \boldsymbol{u}_k - \sum_{i=0}^{I} C_{ik}(s_{ik}), \tag{2}$$

where $\boldsymbol{u}_k := (u_{0k}, u_{1k}, \ldots, u_{Ik})^\top \in \mathbb{R}^{I+1}$ collects the deterministic utilities with $u_{0k} := 0$ (the outside option) and $u_{ik} := v_{ik} - \gamma_k p_i$ for $i \in [I]$, with perturbation function

$$C_{ik}(s_{ik}) = \alpha_{ik}\, s_{ik} \ln s_{ik} \quad \forall i \in [I] \cup \{0\}. \tag{3}$$

We use this equivalent formulation whenever convexity simplifies the analysis, while continuing to interpret the primitives through the generalized rational-inattention lens introduced above.

Normalization and scope. Inspecting the KKT optimality conditions of the convex programming problem (1) shows that only the ratios v_{ik}/γ_k and α_{ik}/γ_k enter the choice probabilities $\boldsymbol{s}_k$; the scale of γ_k cannot be separately identified from shares and prices alone. We therefore adopt the standard normalization $\gamma_k = 1$, which measures the valuations v_{ik} in (1) and the attention weights α_{ik} in (3) in monetary units and, as will become clear in Theorem 1, preserves the semi-closed-form pricing structure. Within-population welfare comparisons are invariant to any common segment-level $\gamma_k > 0$; cross-population welfare comparisons would require recovering γ_k separately.

A further modeling choice concerns price heterogeneity within a segment. Allowing product-specific price coefficients γ_{ik} would yield a richer reduced form, but it would also obscure the clean markup characterization we derive in (5). We retain a common within-segment price coefficient, which trades local flexibility for a pricing formula that is interpretable and usable at deployment scale.

The specification combines four properties that together make it well suited to our setting: (i) product-level substitution heterogeneity that breaks IIA, (ii) LP-based calibration through the KKT conditions, (iii) a convex pricing problem with a transparent markup structure, and (iv) no reliance on a nesting tree or simulation-based random coefficients. Compared with more generic convex demand systems, the generalized rational-inattention lens supplies an economic interpretation of the parameters and, as we show in §3.4, yields a sharper pricing formula.

3.2. Interpretation and Scope

The information-cost parameter α_{ik} plays a dual role. In the RI interpretation, it measures how costly it is for customers in segment k to process information about option i. In the equivalent convex formulation, the same parameter governs the curvature of the perturbation term and therefore the local substitution pattern around that option. The parameters that explain differential attention thus also determine how aggressively each baggage tier can be priced. Both readings are used in the analysis. The RI and MDM representations are mathematically equivalent and supply complementary interpretive lenses on the structural parameters; the pricing and estimation results follow from convexity of the perturbed-utility problem and stand independently of any cognitive reading of rational inattention. The randomized pricing data reject homogeneous substitution and call for product-level heterogeneity; the generalized-RI specification organizes that heterogeneity through a Bregman information cost (§EC.3) and admits an equivalent random-utility representation (the MDM with

exponential marginals of Mishra et al. 2014), so the calibrated α_{ik} are interpretable both as attention weights and as utility-noise scales. Since we observe only segment-level shares, the RI formulation, which commits neither to individual-level utility maximization nor to an i.i.d. taste-shock distribution, is a closer match to the data than a fully specified random-utility model. The paper's contribution is operational: this structure yields tractable estimation, transparent pricing, and a deployment-ready workflow.

3.3. Identification

The segment-level parameters $(\boldsymbol{v}_k, \boldsymbol{\alpha}_k)$ must be recovered from observed prices and shares. For computation, we work directly with the KKT conditions of the convex programming problem (1) and fit $(\boldsymbol{v}_k, \boldsymbol{\alpha}_k)$ by minimizing an ℓ_1 discrepancy between the two sides of those optimality conditions, evaluated at the observed $(\boldsymbol{p}, \boldsymbol{s})$ pairs. This reduces segment-level calibration to a linear programming problem; the explicit formulation and its finite-sample properties are developed in §4.1. Identifiability of the solution to this calibration problem can be characterized through the design matrix induced by the KKT system. Let κ_1 denote the ℓ_1 minimum gain of this design matrix. The parameters are point-identified whenever $\kappa_1 > 0$, and this quantity also controls the statistical error of the LP estimator in §4.1.

A transparent set of sufficient conditions ensures $\kappa_1 > 0$ in practice. In the horizontal interior case, three ingredients are needed: (i) strictly interior observed shares, so that $\ln s_{ik}$ is well defined; (ii) noncollinear price vectors, so that the price columns of the design matrix span the relevant parameter directions; and (iii) a minimum of $L \geq 3$ distinct price levels per segment, a simple counting requirement derived in §4.1. The airline's experimentation platform delivers exactly this type of exogenous variation by randomly assigning price vectors across booking sessions (Liao 2024), and all field experiments in §6.1 use at least four distinct price levels, so the basic identification requirement is satisfied.

3.4. Optimal Pricing: A Semi-Closed-Form Ladder

The generalized-RI specification admits an explicit pricing characterization. We fix a generic segment and suppress the segment subscript for the remainder of this subsection; the feature-based extension is developed in §4. The two results below establish two properties: seller profit is concave in the induced purchase probabilities, so the pricing problem remains convex despite product-level heterogeneity in α; and the optimal price vector admits a semi-closed-form representation indexed by a single scalar $S^\star$, the aggregate purchase probability.

LEMMA 1 (**Profit representation**). *Adopt the normalization $\gamma_k = 1$ and the horizontal constraint structure (so Ω is the simplex with a single dual multiplier μ). View the seller as choosing prices $\boldsymbol{p}$ and the buyer as best-responding via the RI optimization* (1)*; reparameterize the seller's problem in terms of induced*

purchase probabilities $\boldsymbol{s}$ *via the inverse-demand map implied by the consumer's KKT conditions. Define the* net attractiveness *of product* i *in the segment:*

$$w_i := v_i - \alpha_i + \alpha_0.$$

Then the profit as a function of purchase probabilities $\boldsymbol{s}$ *is*

$$\pi(\boldsymbol{s}) := \sum_{i\in[I]} s_i\big(w_i - \alpha_i \ln s_i + \alpha_0 \ln(1 - \sum_{j\in[I]} s_j) - c_i\big).$$

Henceforth in the pricing analysis (§§3–5), $\boldsymbol{s}$ *denotes the* I*-dimensional vector* $(s_1, \ldots, s_I) \in \mathbb{R}^I$*, with the outside-option share* $s_0 := 1 - \sum_{i\in[I]} s_i = 1 - S$ *recovered as a residual; we reserve the* $(I{+}1)$*-dimensional notation* $\boldsymbol{s}_k$ *of* (1) *for the consumer-side formulation.*

The pricing problem maximizes this profit over purchase probabilities:

$$\max_{\boldsymbol{s}\geq 0} \pi(\boldsymbol{s}) \qquad \text{s.t.} \sum_{i\in[I]} s_i \leq 1 \tag{4}$$

Proposition 1 **(Convexity of the pricing problem)**. *The objective in* (4) *is concave in* $\boldsymbol{s}$*. Consequently the revenue-maximization problem is convex.*

Proposition 1 establishes that product-level heterogeneity in $\boldsymbol{\alpha}$ preserves convexity of the pricing problem. The next theorem characterizes the optimizer through a one-dimensional representation in the aggregate share $S^\star$.

Theorem 1 **(Semi-closed-form price ladder)**. *Under* $\alpha_i > 0$ *for* $i \in [I]$ *and* $\alpha_0 > 0$*, the unique interior maximizer* $\boldsymbol{s}^\star$ *of* (4) *satisfies*

$$p_i^\star \;=\; c_i \;+\; \alpha_i \;+\; \alpha_0 \frac{S^\star}{1 - S^\star}, \qquad i \in [I], \tag{5}$$

where $S^\star = \sum_{i\in[I]} s_i^\star$ *is the aggregate purchase probability at the optimum. The optimal prices satisfy*

$$p_1^\star \leq p_2^\star \leq \cdots \leq p_I^\star \quad \Longleftrightarrow \quad c_1{+}\alpha_1 \leq c_2{+}\alpha_2 \leq \cdots \leq c_I{+}\alpha_I,$$

and the ordering is strict at any position where the right-hand sequence is strict.

Corollary 1 **(Parallel-shift comparative statics)**. *Let the intercepts shift by a common additive scalar,* $w_i \to w_i + z$ *for all* $i \in [I]$ *and some* $z \in \mathbb{R}$*. The optimal aggregate purchase probability* $S^\star$ *is strictly increasing in* z*, and*

$$\frac{\partial p_i^\star}{\partial z} \;=\; \frac{\alpha_0}{(1 - S^\star)^2} \frac{\partial S^\star}{\partial z} \;>\; 0, \qquad i \in [I].$$

The common intercept shift therefore translates the entire optimal price ladder in parallel, with the relative price gaps unchanged:

$$p_i^\star - p_j^\star = (c_i - c_j) + (\alpha_i - \alpha_j), \qquad i, j \in [I].$$

The parallel-shift structure, in which a single scalar z translates the entire optimal price ladder while leaving its shape intact, is the structural property that motivates the feature-based pricing rule of §4.3: there, a feature-linear segment shifter plays the role of z, and the analysis for a common additive intercept transfers directly to feature-driven cross-segment variation. The decomposition (5) admits a transparent interpretation: the optimal price equals the effective cost plus a product-specific attention markup α_i, plus a common demand term $\alpha_0 S^\star/(1-S^\star)$ that shifts the entire ladder. The intercepts w_i affect prices only through the common-markup channel, so the ladder's *shape* separates from its *level*: once the attention weights are estimated, the seller solves for a single scalar $S^\star$ per segment.

Corollary 1 records the symmetric demand-side statement: a common intercept shift z translates the ladder in parallel through the same common-markup channel.

3.5. Comparison with Uniform and Propensity-Based Pricing

The RI pricing rule of Theorem 1 can be compared against two operational baselines: *uniform pricing*, in which a single posted price applies to all customers, and *propensity-based pricing*, in which a personalized price scales monotonically with the segment's purchase probability at a reference price. Although the RI rule weakly dominates both baselines in expected profit, the three rules differ systematically in the distribution of posted prices: under substantial attention heterogeneity, the RI rule posts a lower average price than uniform pricing while the propensity rule posts a higher average price. This contrast carries direct welfare and regulatory implications and is documented empirically in §6.2.

The three rules. We compare the three rules on the same indexing: tier i for segment k. (Pooling structure used in estimation is suppressed here; the pricing rules can each be stated at the segment level.) The RI-optimal price specializes Theorem 1 to segment k:

$$p_{i,k}^{\mathrm{RI}} \;=\; c_i + \alpha_{i,k} + \alpha_{0,k}\,\frac{S_k^\star}{1-S_k^\star},$$

where $S_k^\star$ is segment k's aggregate purchase probability under the optimal ladder. The price varies in k through both the segment's attention costs $(\alpha_{i,k}, \alpha_{0,k})$ and its equilibrium aggregate $S_k^\star$, recovered from segment k's full demand curve.

Uniform pricing sets a single tier-i price that does not vary across segments. With $\bar{s}_i(\boldsymbol{p}) = \mathbb{E}_k[s_{i,k}(\boldsymbol{p})]$ denoting the population-aggregated demand for tier i at the common ladder $\boldsymbol{p}$, the uniform ladder solves

$$\boldsymbol{p}^{\,\mathrm{U}} \;=\; \arg\max_{\boldsymbol{p}} \sum_{i\in[I]} p_i\,\bar{s}_i(\boldsymbol{p}), \qquad p_{i,k}^{\mathrm{U}} \equiv p_i^{\mathrm{U}} \quad \text{for all } k.$$

The same ladder is posted to every segment: $p_{i,k}^{\mathrm{U}}$ is constant in k.

Propensity-based pricing observes a per-segment propensity score $\rho_k = S_k(\boldsymbol{p}_0) = \sum_i s_{i,k}(\boldsymbol{p}_0)$ at a baseline ladder $\boldsymbol{p}_0$ and posts

$$p_{i,k}^{\mathrm{Prop}} \;=\; p_{0,i} + (\bar{p}_i - p_{0,i})\,\frac{\rho_k}{\rho_{\max}}, \qquad \rho_{\max} = \sup_{k'} \rho_{k'},$$

with operator-chosen tier ceilings $\bar{p}_i > p_{0,i}$. In practice the baseline ladder is typically set to the prevailing uniform price, $p_{0,i} = p_i^{\mathrm{U}}$, since the uniform schedule is what the operator already has and the propensity score is fit on data collected at that schedule. Under this calibration $p_{i,k}^{\mathrm{Prop}} \geq p_i^{\mathrm{U}}$ for every (i,k), so the propensity rule is structured as a *markup over uniform* rather than a redistribution around it. The dependence on k enters only through the scalar ρ_k: high-propensity segments are charged closer to the ceiling on every tier; low-propensity segments closer to the uniform-price baseline.

By pointwise revenue maximization, the RI rule earns the highest expected profit. This is the textbook ranking that drives operator demand for personalization. The interesting question is what each rule does to the *average price level*, which is what regulators and the press observe.

Mean dominance of propensity prices over RI prices. Our setting has two kinds of customers: *attentive* ones who compare tiers carefully and buy only below their willingness-to-pay, and *inattentive* ones whose buying barely responds to price. This contrast creates a fundamental tension for uniform pricing. To serve inattentive customers, the uniform price is set high, which excludes attentive customers entirely. RI personalization restores the attentive segment at a moderate price, pulling the population average *below* the uniform price, while the propensity rule is structured as a markup over uniform and therefore sits above it on average. The following proposition formalizes this mean-ordering result in the multi-tier setting.

PROPOSITION 2 (**Propensity prices exceed RI prices in mean**). *Let* $\boldsymbol{p}_0 = \boldsymbol{p}^{\mathrm{U}}$ *and* $\bar{p}_i \geq \max_k p_{i,k}^{\mathrm{RI}}$ *for each tier. Then for every tier* i, $\mathbb{E}_k[p_{i,k}^{\mathrm{RI}}] \leq \bar{p}_i$ *and* $\mathbb{E}_k[p_{i,k}^{\mathrm{Prop}}] = p_i^{\mathrm{U}} + (\bar{p}_i - p_i^{\mathrm{U}})\,\mathbb{E}_k[\rho_k]/\rho_{\max}$. *Moreover,* $\mathbb{E}_k[p_{i,k}^{\mathrm{RI}}] \leq \mathbb{E}_k[p_{i,k}^{\mathrm{Prop}}]$ *iff*

$$\frac{\bar{p}_i - \mathbb{E}_k[p_{i,k}^{\mathrm{RI}}]}{\bar{p}_i - p_i^{\mathrm{U}}} \;\geq\; 1 - \frac{\mathbb{E}_k[\rho_k]}{\rho_{\max}}. \tag{M_i}$$

In particular, if $\mathbb{E}_k[p_{i,k}^{\mathrm{RI}}] \leq p_i^{\mathrm{U}}$, *the inequality holds.*

When $\mathbb{E}_k[p_{i,k}^{\mathrm{RI}}] \leq p_i^{\mathrm{U}}$ (the "personalization-lowers-average" condition discussed above), condition (M_i) holds automatically: $(\bar{p}_i - \mathbb{E}_k[p_{i,k}^{\mathrm{RI}}])/(\bar{p}_i - p_i^{\mathrm{U}}) \geq 1$, which exceeds the right-hand side $1 - \mathbb{E}_k[\rho_k]/\rho_{\max} \leq 1$. Proposition 2 characterizes the markup-style operational variant of propensity-based personalization. Symmetric two-sided propensity schedules, in which low-propensity segments are discounted below p_i^{U} and high-propensity segments marked up above it, would replace the ceiling $\bar{p}_i$ with a symmetric bound and yield a different (and in general non-monotone) mean comparison; they fall outside the operational object deployed in our field test.

4. Estimation under Data Sparsity

Calibrating the RI model from sparse experimental data involves three error sources. Section 4.1 develops a KKT-residual LP with a finite-sample bound. Section 4.2 introduces segment pooling to reduce statistical error, at the cost of *merging bias*. Section 4.3 replaces segment intercepts with feature-based terms, trading *approximation bias* for estimability.

4.1. Single-Segment Calibration

For a specified customer segment, our focus is on one population that follows the model. The input data consists of browsing records indexed by customers $c = 1, \ldots, \mathcal{N}$. Each customer c is described by three observed quantities:

- a feature vector $\boldsymbol{x}_{[c]} \in \mathbb{R}^{|\mathcal{J}|}$,
- a price vector $\boldsymbol{p}_{[c]} = (p_{1[c]}, \ldots, p_{I[c]})^\top \in \mathbb{R}^I$,
- a choice vector $\boldsymbol{s}_{[c]} \in \{0,1\}^{I+1}$, a binary indicator of the chosen option.

We assume each customer visits the system once, so the customer index is equivalent to the session index. On an individual level, $\boldsymbol{s}_{[c]}$ is a unit vector $\boldsymbol{e}_i$ if the customer chose option i (including $i = 0$ for no purchase).

There are K unique feature combinations and L unique price combinations observed in the data. We write $\boldsymbol{x}_k$ for the k-th unique feature combination and $\boldsymbol{p}_l = (p_{1l}, \ldots, p_{Il})^\top$ for the l-th unique price combination ($k \in [K]$, $l \in [L]$). We assume observations cover every combination of k and l. Each pair $(\boldsymbol{x}_k, \boldsymbol{p}_l)$ defines an *experiment*, and the empirical market share for experiment (k, l) is

$$\hat{\boldsymbol{s}}_{kl} = \frac{1}{N_{kl}} \sum_{\substack{c:\, \boldsymbol{x}_{[c]} = \boldsymbol{x}_k, \\ \boldsymbol{p}_{[c]} = \boldsymbol{p}_l}} \boldsymbol{s}_{[c]},$$

where N_{kl} is the number of observations in experiment (k, l).

4.1.1. KKT-Based Calibration The first estimation result is that the RI model can be calibrated by solving a linear optimization problem rather than a simulation-based nonlinear likelihood. The key observation is that the KKT system of the RI demand model (1) is linear in the unknown structural parameters once the market shares are observed. For each experiment (k, l), define the active set $\mathcal{B}_{kl} := \{q \in [m] : A_q \hat{\boldsymbol{s}}_{kl} = b_q\}$; by complementary slackness, $\mu_{qkl} = 0$ for $q \notin \mathcal{B}_{kl}$. For each price level $l \in [L]$, the KKT conditions require

$$\begin{gathered}
-\alpha_{0k}(\ln s^\star_{0kl} + 1) + \textstyle\sum_{q=1}^m \mu^\star_{qkl} A_{q0} = 0, \\
v_{ik} - \alpha_{ik}(\ln s^\star_{ikl} + 1) + \textstyle\sum_{q=1}^m \mu^\star_{qkl} A_{qi} = p_{il}, \quad i = 1, \ldots, I; \\
A\boldsymbol{s}^\star_{kl} \le \boldsymbol{b};\ \mu^\star_{qkl} \ge 0,\ q = 1, \ldots, m; \\
\boldsymbol{\mu}^\star_{kl} \cdot (A\boldsymbol{s}^\star_{kl} - \boldsymbol{b}) = \boldsymbol{0}.
\end{gathered}$$

For a fixed segment k, the decision variables are: utility parameters $\boldsymbol{v}_k = (v_{1k}, \ldots, v_{Ik})^\top \in \mathbb{R}^I$, entropy parameters $\boldsymbol{\alpha}_k = (\alpha_{0k}, \alpha_{1k}, \ldots, \alpha_{Ik})^\top \in \mathbb{R}^{I+1}_+$, and dual multipliers $\boldsymbol{\mu}_{kl} = (\mu_{1kl}, \ldots, \mu_{mkl})^\top \in \mathbb{R}^m_+$, one vector per price level $l \in [L]$. Stacking the KKT residuals across all products and price levels, the calibration takes matrix form. Let $M = (I+1)L$ and $N = 2I + 1 + mL$. Define the parameter vector

$$\boldsymbol{\theta}_k := (\boldsymbol{v}_k^\top,\ \boldsymbol{\alpha}_k^\top,\ \boldsymbol{\mu}_{k1}^\top, \ldots,\ \boldsymbol{\mu}_{kL}^\top)^\top \in \mathbb{R}^N, \tag{6}$$

and the design matrix $\hat{\boldsymbol{A}}_k \in \mathbb{R}^{M\times N}$, constructed as follows. For each price level $l \in [L]$, define the diagonal matrix $\hat{D}_{kl} := -\mathrm{diag}\big(1+\ln \hat{s}_{0kl},\ 1+\ln \hat{s}_{1kl},\ \ldots,\ 1+\ln \hat{s}_{Ikl}\big) \in \mathbb{R}^{(I+1)\times(I+1)}$ and the active-constraint coefficient matrix $B_{kl} \in \mathbb{R}^{(I+1)\times m}$ with entries $[B_{kl}]_{iq} = A_{qi} \cdot \mathbf{1}\{q \in \mathcal{B}_{kl}\}$. Let $P := (\mathbf{0},\, I_I)^\top \in \mathbb{R}^{(I+1)\times I}$. Then

$$\hat{\boldsymbol{A}}_k = \begin{pmatrix} P & \hat{D}_{k1} & B_{k1} & \mathbf{0} & \cdots & \mathbf{0} \\ P & \hat{D}_{k2} & \mathbf{0} & B_{k2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ P & \hat{D}_{kL} & \mathbf{0} & \mathbf{0} & \cdots & B_{kL} \end{pmatrix} \in \mathbb{R}^{M\times N}, \tag{7}$$

where the columns are partitioned into the $\boldsymbol{\nu}$-block (P), $\boldsymbol{\alpha}$-block ($\hat{D}_{kl}$), and $\boldsymbol{\mu}$-block (block-diagonal in B_{kl}). The target vector is

$$[\boldsymbol{\nu}]_{(i,l)} := \begin{cases} p_{il} & \text{if } i \in [I], \\ 0 & \text{if } i = 0, \end{cases}$$

for each $l \in [L]$, so that $\boldsymbol{\nu} \in \mathbb{R}^M$. The calibration minimizes the KKT residual:

$$\min_{\boldsymbol{\theta}_k \in \Theta} \|\hat{\boldsymbol{A}}_k\, \boldsymbol{\theta}_k - \boldsymbol{\nu}\|, \tag{8}$$

where $\Theta := \{(\boldsymbol{\nu}_k, \boldsymbol{\alpha}_k, \boldsymbol{\mu}_{k1}, \ldots, \boldsymbol{\mu}_{kL}) \mid \boldsymbol{\alpha}_k \geq \mathbf{0},\ \boldsymbol{\mu}_{kl} \geq \mathbf{0}\ \forall\, l\}$ and $\|\cdot\|$ denotes the ℓ_1 norm; the calibration is therefore a Least Absolute Deviations (LAD) problem, reformulable as a linear programming problem.

Identification. In the horizontal interior case, $m = 1$, so the design matrix has $M = (I+1)L$ rows and $N = 2I+1+L$ columns. A necessary condition for full column rank is $M \geq N$, equivalently $L \geq \lceil 2+1/I \rceil = 3$ for any $I \geq 1$. Identification thus requires $L \geq 3$ distinct price levels with noncollinear price vectors and strictly interior shares, ensuring $\kappa_1(\boldsymbol{A}_k) > 0$. This is the sense in which experimental price variation is essential in our setting: without sufficiently rich randomized prices, the RI parameters cannot be recovered well enough to support pricing.

4.1.2. Error Analysis: Share Estimation The calibration (8) is affected by estimation errors in $\ln \hat{s}_{ikl}$. The following analysis is stated for a fixed (k,l); we omit subscripts for convenience.

Assumption 1 (**Independent choices**). *Within each segment k and price level l, customer choices are independent and identically distributed draws from a multinomial distribution with share vector $\boldsymbol{s}_{kl}$.*

Under Assumption 1, $X_i \sim \mathrm{Binomial}(n, s_i)$ where $\hat{s}_i = X_i/n$ is the empirical share.

Lemma 2 (**Finite-sample log-share accuracy**). *For any $\epsilon > 0$ and $\eta \in (0,1)$, if the number of observations satisfies*

$$n \;\geq\; \frac{\ln\big(2(I+1)/\eta\big)}{[1-(1+\epsilon)e^{-\epsilon}]\,\min_{i\in[I]\cup\{0\}} s_i},$$

then with probability at least $1-\eta$,

$$|\ln \hat{s}_i - \ln s_i| \leq \epsilon \qquad \textit{for all } i \in [I]\cup\{0\}.$$

The required sample size scales inversely with $\min_i s_i$ (rarer options need more data). For small ϵ, the denominator satisfies $1-(1+\epsilon)e^{-\epsilon} = \frac{\epsilon^2}{2}+O(\epsilon^3)$, so the bound simplifies to $n \geq 2\,\ln\big(2(I+1)/\eta\big)/(\epsilon^2\,\min_i s_i)$, making the standard $\sqrt{n}$-rate transparent. Since the true $s_{\min}$ is unobservable, deployment uses the empirical surrogate $\hat{s}_{\min}$, which is consistent under the standard share-concentration regime in which the empirical shares converge uniformly to their population counterparts.

4.1.3. Error Analysis: Parameter Calibration Assume the observation count is sufficiently large that $|\ln(\hat{s}_i) - \ln(s_i)| \leq \epsilon$ globally. We analyze the impact of the share error ϵ on the calibrated parameters for a fixed segment k. The true (noiseless) market shares s_{ikl} define a true design matrix $\boldsymbol{A}_k \in \mathbb{R}^{M\times N}$ by replacing $\hat{s}_{ikl}$ with s_{ikl} in each $\hat{D}_{kl}$. Let $E_k = \hat{\boldsymbol{A}}_k - \boldsymbol{A}_k$; since only the $\boldsymbol{\alpha}$-block is perturbed, E_k inherits a block structure with nonzero entries only in the diagonal positions corresponding to $\hat{D}_{kl} - D_{kl}$. Under $|\ln \hat{s}_{ikl} - \ln s_{ikl}| \leq \epsilon$, every entry of E_k is bounded by ϵ. The natural conditioning measure is the ℓ_1-*minimum gain*: $\kappa_1(\boldsymbol{A}) := \min_{\|\boldsymbol{d}\|_1=1} \|\boldsymbol{A}\,\boldsymbol{d}\|_1$, which is positive when $\boldsymbol{A}$ has full column rank and can be computed via LP. Define the ℓ_1 calibration loss

$$\ell_1(\boldsymbol{\theta};\, \boldsymbol{A}, \boldsymbol{\nu}) := \|\boldsymbol{A}\,\boldsymbol{\theta} - \boldsymbol{\nu}\|_1.$$

Let $\boldsymbol{\theta}_k^\star := \arg\min_{\boldsymbol{\theta}_k\in\Theta} \ell_1(\boldsymbol{\theta}_k; \boldsymbol{A}_k, \boldsymbol{\nu})$ and $\hat{\boldsymbol{\theta}}_k := \arg\min_{\boldsymbol{\theta}_k\in\Theta} \ell_1(\boldsymbol{\theta}_k; \hat{\boldsymbol{A}}_k, \boldsymbol{\nu})$ denote the noiseless and empirical minimizers, respectively, and let $\boldsymbol{\rho}_k := \boldsymbol{\nu} - \boldsymbol{A}_k\boldsymbol{\theta}_k^\star$ be the noiseless residual.

Assumption 2 (**Well-conditioned design**). *The true design matrix $\boldsymbol{A}_k$ has full column rank with $\kappa_1(\boldsymbol{A}_k) > 0$, and the perturbation satisfies $\|E_k\|_{1\to1} < \kappa_1(\boldsymbol{A}_k)$, where $\|E_k\|_{1\to1} := \max_j \sum_i |[E_k]_{ij}|$ is the maximum absolute column sum.*

Theorem 2 (**Finite-sample LP calibration bound**). *Under Assumption 2, the ℓ_1 parameter error satisfies $\|\hat{\boldsymbol{\theta}}_k - \boldsymbol{\theta}_k^\star\|_1 \leq \mathcal{E}_k$, where*

$$\mathcal{E}_k \;:=\; \frac{2\big(\|E_k\|_{1\to1}\,\|\boldsymbol{\alpha}_k^\star\|_1 + \|\boldsymbol{\rho}_k\|_1\big)}{\kappa_1(\boldsymbol{A}_k) - \|E_k\|_{1\to1}}. \tag{9}$$

Since E_k has nonzero entries only in the $\boldsymbol{\alpha}$-block columns, $E_k\boldsymbol{\theta}_k = E_k\,\iota_\alpha(\boldsymbol{\alpha}_k)$ for every $\boldsymbol{\theta}_k \in \mathbb{R}^N$ of the form (6), where $\iota_\alpha : \mathbb{R}^{I+1} \to \mathbb{R}^N$ denotes the zero-padded embedding that places $\boldsymbol{\alpha}_k$ in the $\boldsymbol{\alpha}$-block and zeros elsewhere. Consequently $\|E_k\boldsymbol{\theta}_k\|_1 \leq \|E_k\|_{1\to1}\|\boldsymbol{\alpha}_k\|_1$; moreover, each α_i-column of E_k has exactly L perturbed entries bounded by ϵ, so $\|E_k\|_{1\to1} \leq L\,\epsilon$, yielding

$$\mathcal{E}_k \;\leq\; \frac{2\big(L\,\epsilon\,\|\boldsymbol{\alpha}_k^\star\|_1 + \|\boldsymbol{\rho}_k\|_1\big)}{\kappa_1(\boldsymbol{A}_k) - L\,\epsilon}. \tag{10}$$

Theorem 2 characterizes the LP estimator: the error is small when the design is well-conditioned (large $\kappa_1(\boldsymbol{A}_k)$), the information-cost parameters are bounded, and the share estimates are precise. The factor of 2 originates from bounding the numerator via the triangle inequality on the ℓ_1 residual perturbation; it is tight when the perturbation and the residual reinforce rather than cancel.

4.2. Customer Segmentation and the Merging-Bias Trade-off

The single-segment analysis raises the question of how finely customers should be segmented. Finer segmentation captures richer heterogeneity but distributes the data thinner across cells; coarser segmentation pools more observations per cell but blurs genuine demand differences. The trade-off is the central statistical design choice and directly determines downstream pricing quality.

We specify $\mathcal{J}_S \subset \mathcal{J}$ as the set of segmentation features, yielding $|G_{\mathcal{J}_S}|$ groups of parameters. For each fine segment $k \in g \in G_{\mathcal{J}_S}$, $v_{ik} = v_i^{(g)}$, $\alpha_{ik} = \alpha_i^{(g)}$, and $\mu_{qkl} = \mu_{ql}^{(g)}$.

Consider two segmentation schemes: the *fine* scheme using all features $\mathcal{J}$, which assigns each of the $K = |G_{\mathcal{J}}|$ unique feature combinations its own parameter, and the *coarse* scheme using $\mathcal{J}_S \subset \mathcal{J}$, which pools segments into $|G_{\mathcal{J}_S}|$ groups. Each coarse group $g \in G_{\mathcal{J}_S}$ contains r_g fine segments, and we write $r_{\max} := \max_g r_g$. The fine scheme estimates separate parameters for each segment k but uses fewer observations per segment; the coarse scheme pools observations within each group g, gaining statistical efficiency at the cost of imposing parameter homogeneity.

Merged shares. For each coarse group $g \in G_{\mathcal{J}_S}$, product i, and price level l, the pooled empirical share is

$$\hat{s}_{igl} = \frac{1}{N_{gl}} \sum_{\substack{c:\, \boldsymbol{x}_{[c]} \in g, \\ \boldsymbol{p}_{[c]} = \boldsymbol{p}_l}} [\boldsymbol{s}_{[c]}]_i, \qquad N_{gl} = \sum_{k \in g} N_{kl}, \tag{11}$$

which is equivalently a weighted average of the fine-segment empirical shares: $\hat{s}_{igl} = \sum_{k \in g} (N_{kl}/N_{gl})\, \hat{s}_{ikl}$.

Group design matrix and calibration. For each group g, define the group design matrix $\hat{\boldsymbol{A}}_g \in \mathbb{R}^{M \times N}$ with the same block structure as (7), replacing fine-segment shares with merged shares (11). The group calibration solves $\min_{\boldsymbol{\theta}_g \in \Theta} \|\hat{\boldsymbol{A}}_g \boldsymbol{\theta}_g - \boldsymbol{\nu}\|_1$. Let $\boldsymbol{A}_g$ denote the noiseless counterpart (replacing $\hat{s}_{igl}$ with true merged shares $\bar{s}_{igl}$), $\boldsymbol{\theta}_g^\star$ the noiseless minimizer, and $\boldsymbol{\rho}_g := \boldsymbol{\nu} - \boldsymbol{A}_g \boldsymbol{\theta}_g^\star$ the noiseless residual (nonzero when within-group shares are heterogeneous). Write $\kappa_{1,g} := \kappa_1(\boldsymbol{A}_g)$.

LEMMA 3 **(Bias from share merging)**. *For any fine segment $k \in g$, assume the model is correctly specified at the fine-segment level, i.e., $\boldsymbol{A}_k \boldsymbol{\theta}_k^\star = \boldsymbol{\nu}$, and that the group design $\boldsymbol{A}_g$ satisfies $\kappa_{1,g} := \kappa_1(\boldsymbol{A}_g) > 0$. Then the bias from share merging under ℓ_1 calibration satisfies*

$$\|\boldsymbol{\theta}_g^\star - \boldsymbol{\theta}_k^\star\|_1 \le \frac{\|(\boldsymbol{A}_g - \boldsymbol{A}_k)\, \boldsymbol{\theta}_k^\star + \boldsymbol{\rho}_g\|_1}{\kappa_{1,g}}.$$

The merging bias is driven by within-group share heterogeneity $\ln \bar{s}_{igl} - \ln s_{ikl}$: if all fine segments in group g have identical true shares, $\boldsymbol{A}_g = \boldsymbol{A}_k$ and $\boldsymbol{\rho}_g = \boldsymbol{0}$, so the bias vanishes. Note that $\boldsymbol{\rho}_g$ is computable from group-level data alone. When the fine-segment level has nonzero noiseless residual $\boldsymbol{\rho}_k = \boldsymbol{\nu} - \boldsymbol{A}_k \boldsymbol{\theta}_k^\star \neq \boldsymbol{0}$ (so the model is not exactly specified at the fine level), the same derivation gives the more general bound $\|\boldsymbol{\theta}_g^\star - \boldsymbol{\theta}_k^\star\|_1 \le \|(\boldsymbol{A}_g - \boldsymbol{A}_k)\boldsymbol{\theta}_k^\star + \boldsymbol{\rho}_g - \boldsymbol{\rho}_k\|_1 / \kappa_{1,g}$; the displayed result is the specialization to $\boldsymbol{\rho}_k = \boldsymbol{0}$.

Combining the merging bias with the estimation error yields a bias–variance decomposition. For the fine scheme, $\mathcal{E}_k$ is evaluated at sample size N_k; for the coarse scheme, $\mathcal{E}_g$ denotes the bound (9) applied to the group calibration with design matrix $\boldsymbol{A}_g$ and pooled sample size $N_g = \sum_{k \in g} N_k$.

THEOREM 3 **(Bias–variance trade-off under segment pooling)**. *Assume* $\kappa_{1,g} > 0$ *(Assumption 2 applied to the group design* $\boldsymbol{A}_g$*). For a fine segment* k *belonging to coarse group* g*, the total parameter error of the coarse scheme under* ℓ_1 *calibration decomposes as*

$$\|\hat{\boldsymbol{\theta}}_g - \boldsymbol{\theta}_k^\star\|_1 \le \underbrace{\frac{\|(\boldsymbol{A}_g - \boldsymbol{A}_k)\,\boldsymbol{\theta}_k^\star + \boldsymbol{\rho}_g\|_1}{\kappa_{1,g}}}_{\text{merging bias}} + \underbrace{\mathcal{E}_g.}_{\text{estimation error}}$$

Theorem 3 states the bias–variance trade-off explicitly: coarse pooling is preferred when the reduction in estimation error outweighs the distortion from forcing neighboring segments to share a parameter vector. In data-limited regimes the prescription is to start with coarse pooling and refine only when additional heterogeneity dominates the statistical penalty. Since revenue loss scales quadratically in $\|\Delta\boldsymbol{\theta}\|_1$ (Theorem 5), the same trade-off governs revenue loss.

4.3. Feature-Based Approximation

When $|\mathcal{J}_S|$ is large, $|G_{\mathcal{J}_S}|$ grows combinatorially. We therefore introduce a second level of pooling that is especially relevant for deployment. Select $\mathcal{J}_L \subset \mathcal{J}_S$ whose effects on utility are approximated linearly, and define groups only by $\mathcal{J}_S \setminus \mathcal{J}_L$. This restriction is what ultimately makes feature-based pricing possible: some booking characteristics change the overall level of demand but need not induce a completely separate RI calibration problem. Within each group $h \in G_{\mathcal{J}_S \setminus \mathcal{J}_L}$, the parameters are

$$v_{ik} = v_i^{(h)} + \boldsymbol{\beta}_{\mathcal{J}_L}^{(h)} \cdot \boldsymbol{x}_{\mathcal{J}_L,k}, \qquad \alpha_{ik} = \alpha_i^{(h)}, \qquad \mu_{qkl} = \mu_{ql}^{(h)},$$

where $v_i^{(h)}$ is a group-specific base utility and $\boldsymbol{\beta}_{\mathcal{J}_L}^{(h)} \in \mathbb{R}^{|\mathcal{J}_L|}$ captures the marginal feature effects on utility. The product-independent $\boldsymbol{\beta}$ shifts all utilities equally, producing a parallel-shift pricing structure (§EC.5 of the e-companion) and preserving LP calibration. This trades flexibility for efficiency: the linear structure is exact when feature effects are additive.

h-level calibration. For each $h \in G_{\mathcal{J}_S \setminus \mathcal{J}_L}$, let $G_h := \{g \in G_{\mathcal{J}_S} : g \subset h\}$ denote the $\mathcal{J}_S$-groups within h. The h-level calibration jointly fits all $g \in G_h$, optimizing over the parameter

$$\boldsymbol{\theta}_h^{\text{lin}} := (\boldsymbol{v}^{(h)\top}, \boldsymbol{\beta}_{\mathcal{J}_L}^{(h)\top}, \boldsymbol{\alpha}^{(h)\top}, \boldsymbol{\mu}^{(h)\top})^\top,$$

where $\boldsymbol{v}^{(h)} \in \mathbb{R}^I$ and $\boldsymbol{\alpha}^{(h)} \in \mathbb{R}^{I+1}$ are shared across all $g \in G_h$, $\boldsymbol{\beta}_{\mathcal{J}_L}^{(h)} \in \mathbb{R}^{|\mathcal{J}_L|}$ captures the feature effects on utility, and $\boldsymbol{\mu}^{(h)} := (\boldsymbol{\mu}^{(g_1)\top}, \ldots, \boldsymbol{\mu}^{(g_{|G_h|})\top})^\top$ concatenates the g-level Lagrange multipliers, which remain group-specific since each g has its own active constraint sets $\mathcal{B}_{gl}$. The h-level design matrix $\hat{\boldsymbol{A}}_h^{\text{lin}}$ stacks the modified design matrices across all $g \in G_h$ (see §EC.6 of the e-companion for the explicit block structure). The shared parameter count drops from $(2I+1) \cdot |G_{\mathcal{J}_S}|$ to $(2I+1+|\mathcal{J}_L|) \cdot |G_{\mathcal{J}_S \setminus \mathcal{J}_L}|$.

Induced g-level parameter. For each $g \in G_h$, define

$$\boldsymbol{\theta}_{g|h} := \big(\boldsymbol{\nu}^{(h)} + (\boldsymbol{x}_g^\top \boldsymbol{\beta}_{\mathcal{J}_L}^{(h)})\,\mathbf{1}_I,\ \boldsymbol{\alpha}^{(h)},\ \boldsymbol{\mu}^{(g)}\big) \in \mathbb{R}^N. \tag{12}$$

Let $\tilde{\boldsymbol{\theta}}^\star_{g|h}$ denote the induced g-level parameter from the noiseless h-level minimizer, and $\hat{\boldsymbol{\theta}}_{g|h}$ the empirical counterpart. The subscript $g|h$ reads "g-level induced from h-group calibration."

LEMMA 4 **(Bias from feature-based approximation)**. *Assume $\kappa_{1,g} > 0$. For any $g \in G_h$, the bias from the h-level parameterization under ℓ_1 calibration satisfies*

$$\|\tilde{\boldsymbol{\theta}}^\star_{g|h} - \boldsymbol{\theta}^\star_g\|_1 \ \le\ \frac{\|\boldsymbol{\rho}_g - \tilde{\boldsymbol{\rho}}_{g|h}\|_1}{\kappa_{1,g}},$$

where $\tilde{\boldsymbol{\rho}}_{g|h} := \boldsymbol{\nu} - \boldsymbol{A}_g\,\tilde{\boldsymbol{\theta}}^\star_{g|h}$ is the g-level noiseless residual of the induced parameter, and $\boldsymbol{\rho}_g := \boldsymbol{\nu} - \boldsymbol{A}_g\,\boldsymbol{\theta}^\star_g$.

When the linear model is exact and $\boldsymbol{\alpha}^{(g)}$ is homogeneous across G_h, the bias vanishes. The restriction $\alpha_{ik} = \alpha_i^{(h)}$ is a parsimony choice: feature-dependent α_{ik} would add $(I{+}1)\,|\mathcal{J}_L|$ coefficients while preserving the LP structure, but the segmentation step already captures α-heterogeneity across h-groups.

Combining the three error sources requires the following construction. Let $\mathcal{E}_{g|h}$ denote the bound (9) applied to the h-level calibration with design matrix $\boldsymbol{A}_h^{\mathrm{lin}}$ and pooled sample size $N_h = \sum_{g\in G_h} N_g$. The induced g-level parameter $\boldsymbol{\theta}_{g|h}$ is the image of the stacked restricted parameter $\boldsymbol{\theta}_h^{\mathrm{lin}}$ under a linear map $L_g : \mathbb{R}^{\dim(\boldsymbol{\theta}_h^{\mathrm{lin}})} \to \mathbb{R}^N$: from (12), the $\boldsymbol{\nu}$-block of $\boldsymbol{\theta}_{g|h}$ equals $\boldsymbol{\nu}^{(h)} + (\boldsymbol{x}_g^\top \boldsymbol{\beta}_{\mathcal{J}_L}^{(h)})\mathbf{1}_I$ (a linear combination of $\boldsymbol{\nu}^{(h)}$ and $\boldsymbol{\beta}_{\mathcal{J}_L}^{(h)}$), the $\boldsymbol{\alpha}$-block selects $\boldsymbol{\alpha}^{(h)}$, and the $\boldsymbol{\mu}$-block selects $\boldsymbol{\mu}^{(g)}$ from the stacked $(\boldsymbol{\mu}^{(g_1)}, \ldots, \boldsymbol{\mu}^{(g_{|G_h|})})$. The induced parameter is therefore not a sub-vector of $\boldsymbol{\theta}_h^{\mathrm{lin}}$, and the per-$g$ estimation error satisfies $\|\hat{\boldsymbol{\theta}}_{g|h} - \tilde{\boldsymbol{\theta}}^\star_{g|h}\|_1 = \|L_g(\hat{\boldsymbol{\theta}}_h^{\mathrm{lin}} - \tilde{\boldsymbol{\theta}}_h^{\mathrm{lin},\star})\|_1 \le \eta_{g|h}\,\|\hat{\boldsymbol{\theta}}_h^{\mathrm{lin}} - \tilde{\boldsymbol{\theta}}_h^{\mathrm{lin},\star}\|_1 \le \eta_{g|h}\,\mathcal{E}_{g|h}$, where

$$\eta_{g|h} \ :=\ \|L_g\|_{1\to 1} \ \le\ \max\big(1,\ I\,\|\boldsymbol{x}_g\|_\infty\big) \tag{13}$$

is the $\ell_1 \to \ell_1$ operator norm of L_g, computed as the maximum column ℓ_1-sum of L_g: $\boldsymbol{\nu}$- and $\boldsymbol{\alpha}$-columns each contribute 1, each $\boldsymbol{\beta}_{\mathcal{J}_L}$-column contributes $\sum_{i\in[I]} |x_{g,j}| \le I\|\boldsymbol{x}_g\|_\infty$ (it injects $x_{g,j}$ into every one of the I products' $\boldsymbol{\nu}$-slots), and each $\boldsymbol{\mu}^{(g)}$-column contributes 1 (other groups' columns contribute 0).

THEOREM 4 **(Three-way error decomposition under feature pooling)**. *Assume $\kappa_{1,g} > 0$ (Assumption 2 applied to the group design $\boldsymbol{A}_g$). For a fine segment k belonging to $\mathcal{J}_S$-group g within h-group, the total parameter error of the feature-based scheme under ℓ_1 calibration decomposes as*

$$\|\hat{\boldsymbol{\theta}}_{g|h} - \boldsymbol{\theta}^\star_k\|_1 \le \underbrace{\frac{\|(\boldsymbol{A}_g - \boldsymbol{A}_k)\,\boldsymbol{\theta}^\star_k + \boldsymbol{\rho}_g\|_1}{\kappa_{1,g}}}_{\text{merging bias}} + \underbrace{\frac{\|\boldsymbol{\rho}_g - \tilde{\boldsymbol{\rho}}_{g|h}\|_1}{\kappa_{1,g}}}_{\text{linear approx. bias}} + \underbrace{\eta_{g|h}\cdot\mathcal{E}_{g|h}}_{\text{est. error}},$$

where $\eta_{g|h} := \|L_g\|_{1\to 1} \le \max(1, I\|\boldsymbol{x}_g\|_\infty)$ is the $\ell_1 \to \ell_1$ operator norm of the induced linear map L_g defined by (12); see (13).

Theorem 4 completes the decomposition. Enlarging $\mathcal{J}_L$ pools more groups into each h-group, reducing the estimation error at the cost of additional linear-approximation bias; when $\mathcal{J}_L = \emptyset$, the bound reduces to Theorem 3. The result provides a formal rationale for the feature-based pricing rule: a controlled amount of local flexibility is traded for substantially improved statistical efficiency in the regime of many booking states with limited per-state traffic.

Feature-group pricing ladder. The semi-closed-form ladder of Theorem 1 applies to each g-group within an h-group under the parameters $(\boldsymbol{v}^{(h)}, \boldsymbol{\alpha}^{(h)}, \boldsymbol{c}^{(h)})$ shared across $g \in G_h$, with intercepts augmented by the segment shifter $z_g := \boldsymbol{\beta}^{(h)\top} \boldsymbol{x}_g$. Writing $w_i^{(h)} := v_i^{(h)} - \alpha_i^{(h)} + \alpha_0^{(h)}$ for net attractiveness, the g-level optimal prices read

$$p_{i,g}^{\star} = c_i^{(h)} + \alpha_i^{(h)} + \alpha_0^{(h)} \frac{S_g^{\star}}{1 - S_g^{\star}}, \qquad i \in [I],$$

where $S_g^{\star}$ solves the segment-level first-order conditions with intercepts $w_i^{(h)} + z_g$. Because z_g enters the utilities as a common additive term across products, Corollary 1 applies with $z = z_g$: varying the segment features shifts the entire g-level price ladder in parallel, with per-product derivative $\partial p_{i,g}^{\star} / \partial z_g > 0$ common across i. The quantitative sensitivity bound $\partial p_{i,g}^{\star} / \partial z_g \le 1/(2 - S_g^{\star}) \le 1$ is derived in §EC.5 and underlies the feature-based pricing rule (§5.2).

From bounds to practice. The bias–variance trade-off in Theorems 3 and 4 is operationalized by a split-sample cross-validation procedure that scores each candidate configuration by its share-prediction error on held-out data, yielding a data-driven criterion for partitioning features into segmentation and linear components. The procedure is developed in §EC.2 of the e-companion; §EC.7 reports that it attains 96%–99% of oracle revenue in synthetic experiments.

5. Decision Robustness and Pricing Rules

The parameter-error bounds from §4 translate into revenue-loss guarantees that, in turn, govern the relative performance of three implementable pricing rules: segment-based (SEG), feature-based (FEA), and propensity-based (PCP). These rules span the data–complexity trade-off, and the analysis below identifies when the simpler feature-based rule remains competitive with exact segment-based pricing. The same rules are deployed in the field experiment of §6.1.

5.1. Revenue Loss under Estimation Error

Statistical uncertainty in the calibrated parameters propagates into revenue loss in a way that admits a clean quadratic bound. Theorem 5 converts an ℓ_1 bound on parameter error into a quadratic bound on the revenue forgone by pricing with the estimated model rather than the true one. The argument rests on three regularity conditions that ensure uniform curvature and smoothness of the profit function.

Assumption 3 (**Positive curvature**). *There exists $\underline{\alpha} > 0$ such that $\alpha_i \ge \underline{\alpha}$ for all $i \in [I]$, and the true parameter $\boldsymbol{\theta}^{\star}$ lies in a compact set $\boldsymbol{\Theta}$ on which $\alpha_i \ge \underline{\alpha}$ for $i \in [I]$ and $\alpha_0 > \underline{\alpha}$.*

ASSUMPTION 4 **(Bounded primitive utilities)**. *There exists* $B < \infty$ *such that* $\max_i \left| w_i + \boldsymbol{\beta}^\top \boldsymbol{x} - c_i \right| \le B$.

ASSUMPTION 5 **(Uniform interior solution)**. *There exist constants* $\underline{s} > 0$ *and* $\delta > 0$ *such that, for every* $\boldsymbol{\theta} \in \boldsymbol{\Theta}$*, the unique optimizer* $\boldsymbol{s}^\star(\boldsymbol{\theta})$ *of the pricing problem satisfies*

$$s_i^\star(\boldsymbol{\theta}) \ge \underline{s} \quad \textit{for all } i \in [I], \qquad 1 - \sum_{i \in [I]} s_i^\star(\boldsymbol{\theta}) \ge \delta.$$

Assumptions 3–5 provide the regularity needed for a uniform robustness guarantee. Assumption 5 is a technical condition used only in the revenue-loss analysis: it rules out sequences of parameters whose optimal solutions approach the boundary of the simplex, where curvature constants can explode. The e-companion establishes pointwise interiority and continuity of $\boldsymbol{s}^\star(\boldsymbol{\theta})$ under Assumptions 3–4; Assumption 5 strengthens this to a uniform statement over the compact parameter set $\boldsymbol{\Theta}$. Under these assumptions, uniform curvature and smoothness constants $\mu, L_\pi > 0$ satisfying $\mu I \preceq -\nabla^2_{ss}\pi(\boldsymbol{s}^\star;\boldsymbol{\theta}) \preceq L_\pi I$ exist; the subscript π distinguishes the profit-function smoothness constant from the number of price levels L used in §4. Define the mixed-norm Jacobian bound $G_1 := \|J\|_{1\to 2} = \max_j \|J_{\cdot,j}\|_2$, where $J(\boldsymbol{s};\boldsymbol{\theta}) := \frac{\partial}{\partial \boldsymbol{\theta}^\top} \nabla_{\boldsymbol{s}} \pi(\boldsymbol{s};\boldsymbol{\theta})$.

THEOREM 5 **(Quadratic revenue loss bound)**. *Under Assumptions 3–5, with* $\Delta\boldsymbol{\theta} := \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^\star$*,*

$$\pi\big(\boldsymbol{s}^\star(\boldsymbol{\theta}^\star);\boldsymbol{\theta}^\star\big) - \pi\big(\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}});\boldsymbol{\theta}^\star\big) \;\le\; \frac{L_\pi\, G_1^2}{2\,\mu^2}\, \|\Delta\boldsymbol{\theta}\|_1^2. \tag{14}$$

The parameter $\boldsymbol{\theta}$ *here is the structural pricing parameter* $(\boldsymbol{w}, \boldsymbol{\alpha}, \boldsymbol{c})$ *(and feature-effect* $\boldsymbol{\beta}$*, when applicable) entering* $\pi(\boldsymbol{s};\boldsymbol{\theta})$ *via Lemma 1, with* $w_i := v_i - \alpha_i + \alpha_0$*. The dual multipliers* $\boldsymbol{\mu}_{kl}$ *from the calibration parameterization (6) do not appear in* $\pi(\boldsymbol{s};\cdot)$ *after KKT substitution and contribute zero columns to the Jacobian* J*. An upstream calibration bound* $\|\Delta\boldsymbol{\theta}^{\text{full}}\|_1 \le \mathcal{E}$ *on the* $(\boldsymbol{v}, \boldsymbol{\alpha}, \boldsymbol{\mu})$*-parameterization (Theorems 3 and 4) propagates to (14) via the identity* $\Delta w_i = \Delta v_i - \Delta\alpha_i + \Delta\alpha_0$*, which implies* $\|\Delta(\boldsymbol{w}, \boldsymbol{\alpha}, \boldsymbol{c})\|_1 \le (I{+}1)\, \|\Delta\boldsymbol{\theta}^{\text{full}}\|_1$*. The constant* $I{+}1$ *is the* $\ell_1 \to \ell_1$ *operator norm of the linear map* $(\boldsymbol{v}, \boldsymbol{\alpha}) \mapsto (\boldsymbol{w}, \boldsymbol{\alpha})$*, attained on the* α_0*-column. The resulting bound is* $\frac{L_\pi G_1^2}{2\mu^2}\, (I{+}1)^2\, \mathcal{E}^2$*, preserving the quadratic dependence on* $\mathcal{E}$ *up to an* I*-dependent constant.*

Substituting $\|\Delta\boldsymbol{\theta}\|_1 \le \mathcal{E}$ from Theorems 3 or 4 into (14) yields Revenue loss $\le (L_\pi\, G_1^2/2\mu^2)\, \mathcal{E}^2$, linking calibration error to revenue loss. The ℓ_1-based bound avoids the conservative $\|\Delta\boldsymbol{\theta}\|_2 \le \|\Delta\boldsymbol{\theta}\|_1$ inequality and uses the smaller constant $G_1 \le G$. The constants μ, L_π, G_1 are computable from the calibrated parameters, so the bound is implementable for deployment-risk assessment.

5.2. Segment-Based versus Feature-Based Pricing

The estimation framework of §4 induces two natural pricing rules that span the bias–variance trade-off: an exact but data-intensive segment-level rule, and a feature-based rule that is easier to deploy when per-segment data are thin.

Segment-based pricing (SEG). SEG solves the pricing problem (4) independently for each g-group, computing the exact equilibrium aggregate $S_g^\star$ and obtaining the optimal prices from the semi-closed form (5). The utility shift $z_g = \boldsymbol{\beta}^{(h)\top}\boldsymbol{x}_g$ determines a distinct equilibrium for each g-group, yielding per-group price ladders that capture the nonlinear dependence of optimal prices on segment features.

Feature-based pricing (FEA). FEA is a simplified pricing rule motivated by the parallel-shift structure of the RI solution. For each g-group within an h-group,

$$\boldsymbol{p}_{\mathrm{FEA},g} := \boldsymbol{p}^{\dagger}_{\mathrm{FEA}} + \big(\boldsymbol{\beta}^{(h)\top}\boldsymbol{x}_g\big)\mathbf{1},$$

where $\boldsymbol{p}^{\dagger}_{\mathrm{FEA}}$ is the base price ladder computed at the default feature vector. The equality defines the rule; it is not an exact optimality identity. The exact RI-optimal prices shift in parallel with z_g, but with slope strictly below one. By calibrating a single h-level model that pools all g-groups within each h-group, FEA reduces the effective number of free parameters while retaining group-level price differentiation through $\boldsymbol{\beta}$.

The FEA–SEG gap equals the cost of the parallel-shift approximation. By Corollary EC.1, the exact derivative satisfies $\partial p_i^\star/\partial z \in (0,\, 1/(2 - S_g^\star)] \subset (0, 1]$, so FEA's unit-slope correction systematically overshoots; the approximation tightens as $S^\star \to 1$. FEA trades approximation bias for lower estimation variance (Theorem 4); SEG closes the gap when per-group data are sufficient. An empirical comparison of FEA and SEG by sample size in the field deployment is reported in §EC.8 of the e-companion.

5.3. A Deployable Propensity-Based Rule

The third rule is the deployable counterpart of the propensity-based pricing of §3.5: it replaces the structural propensity score $\rho_k = S_k(\boldsymbol{p}_0)$ with a machine-learning predictor $\hat{q}$ of aggregate purchase probability, and the continuous markup with a one-dimensional threshold adjustment of posted prices. It holds the demand-prediction input flexible and varies only the pricing rule. We tag this deployed arm *PCP* (predicted choice probability) in the field experiment of §6.1, where it is the personalized-pricing benchmark against which the attention-aware rules SEG and FEA are evaluated.

Setup. Let $\mathcal{D} = \{(\boldsymbol{x}^{(t)}, \boldsymbol{p}^{(t)}, y^{(t)})\}_{t=1}^{T}$ denote a training panel of T booking sessions, where $\boldsymbol{x}^{(t)} \in \mathcal{X}$ is the observed feature vector, $\boldsymbol{p}^{(t)} \in \mathbb{R}_+^I$ the posted price vector across the I product tiers, and $y^{(t)} \in \{0\} \cup [I]$ the realized choice. Let $\mathcal{F}$ be a hypothesis class of probability predictors (e.g., regularized binary logit, gradient-boosted trees, or neural networks) that map $(\boldsymbol{x}, \boldsymbol{p})$ to the aggregate purchase probability $\Pr(y \neq 0 \mid \boldsymbol{x}, \boldsymbol{p})$.

Algorithm. Given an incoming session with features $\boldsymbol{x}$ and an existing posted-price vector $\boldsymbol{p}^0$, PCP pricing applies a single, common shift to all product tiers and selects the deployed price vector $\boldsymbol{p}^{\mathrm{PCP}}(\boldsymbol{x})$ in three steps:

1. **Probability estimation.** Solve, offline,

$$\hat{q} \in \arg\min_{q \in \mathcal{F}} \frac{1}{T}\sum_{t=1}^{T} \ell\big(q(\boldsymbol{x}^{(t)}, \boldsymbol{p}^{(t)}),\, \mathbf{1}\{y^{(t)} \neq 0\}\big) + \Omega(q), \tag{15}$$

where ℓ is a proper scoring loss (e.g., cross-entropy) and Ω a regularizer. The estimate $\hat{q}(\boldsymbol{x}, \boldsymbol{p})$ approximates the aggregate purchase probability $\Pr(y \neq 0 \mid \boldsymbol{x}, \boldsymbol{p})$, pooling across all I tiers.

2. **Reference target.** Fix a target purchase probability $\bar{q}^\star \in (0,1)$ and a tolerance band $[\underline{q}, \overline{q}]$ with $\underline{q} < \bar{q}^\star < \overline{q}$. A standard choice is $\bar{q}^\star = \mathbb{E}_{\boldsymbol{x}}[\hat{q}(\boldsymbol{x}, \boldsymbol{p}^0)]$, the average predicted aggregate purchase probability at the current posted-price vector.
3. **Threshold price adjustment.** With step size $\delta > 0$,

$$\boldsymbol{p}^{\mathrm{PCP}}(\boldsymbol{x}) \;:=\; \boldsymbol{p}^0 \;+\; \delta \cdot \operatorname{sgn}\big(\hat{q}(\boldsymbol{x}, \boldsymbol{p}^0) - \bar{q}^\star\big) \cdot \mathbf{1}\big\{\hat{q}(\boldsymbol{x}, \boldsymbol{p}^0) \notin [\underline{q}, \overline{q}]\big\} \cdot \mathbf{1}, \tag{16}$$

where $\mathbf{1} \in \mathbb{R}^I$ is the all-ones vector, so that all I posted prices move up by δ together when predicted aggregate demand exceeds the upper tolerance, down by δ together when it falls below the lower tolerance, and are unchanged otherwise.

The hyperparameters $(\bar{q}^\star, \underline{q}, \overline{q}, \delta)$ and the predictor family $\mathcal{F}$ are tuned offline. The deployed update (16) is closed form and requires a single $\hat{q}$ evaluation per arriving session. By construction the rule preserves the baseline tier ladder $\boldsymbol{p}^0$ and varies it across sessions only through a scalar markup applied uniformly to all tiers.

Relation to the structural rule. PCP shares the directional logic of attention-aware pricing (raise the price where predicted demand is robust, lower it where it is fragile): $\hat{q}(\boldsymbol{x}, \boldsymbol{p}^0)$ plays the operational role of the structural aggregate share $S^\star(\boldsymbol{x}; \boldsymbol{\theta})$ in the closed-form ladder (5), and the sign rule in (16) matches the comparative statics $\partial p_i^\star / \partial z > 0$ of Corollary EC.1 at $z = \boldsymbol{\beta}^\top \boldsymbol{x}$, applied uniformly across tiers. It implements this, however, through a hand-tuned scalar perturbation rather than by solving the seller's convex program: as a black-box surrogate it preserves the direction of the optimal response but not its magnitude, and recovers neither the product-specific attention markup α_i of Theorem 1 nor cross-tier substitution. The predictor family $\mathcal{F}$ may in return be arbitrarily expressive (nonlinear interactions and individual-level signals refreshed at the session level), so PCP buys finer demand prediction at the cost of coarser pricing, whereas FEA and SEG accept a parsimonious demand specification to solve the pricing problem in closed form.

6. Field Deployment in Airline Ancillaries

The framework is validated on two fronts. A synthetic study (§EC.9) confirms that the heterogeneous-attention specification matches MNL when substitution is homogeneous ($\sigma_\alpha = 0$) and delivers substantial gains otherwise: under finite-sample noise at $n = 2{,}000$, MNL's revenue loss exceeds the heterogeneous-attention model's by roughly an order of magnitude at $\sigma_\alpha = 0.5$ and by more than 200× at $\sigma_\alpha = 1.5$, while MNL's loss plateaus at a positive misspecification floor as the sample grows. Given the tier-elasticity gradient documented in §2, this synthetic evidence indicates that the heterogeneous-attention specification is the appropriate basis for deployment. The field experiment that follows evaluates the revenue impact of the three pricing rules in a live booking environment.

6.1. Field Experiment: Design and Randomization

The pricing framework is deployed in collaboration with the Airline of §2.[1] The main experiment covers 48 origin–destination pairs spanning 24 routes and ran from August 12 to September 30, 2025. Randomization is implemented at the booking-session level across four arms: approximately 60% of sessions to feature-based pricing (FEA), 30% to segment-based pricing (SEG), 5% to the PCP rule of §5.3, and 5% to a uniform-pricing control. FEA and SEG receive the majority of the traffic as the two candidate revenue-improvement rules, with the PCP and uniform-control arms serving as benchmarks for personalized and unpersonalized pricing, respectively.

The PCP benchmark arm. The PCP arm is the deployed propensity-based rule of §5.3: its base price and shared feature shift $\sum_{j\in\mathcal{J}}\beta_j x_{kj}$ (a single coefficient β_j across all baggage tiers, no tier-specific β_{ij}) play the roles of the uniform baseline p_i^{U} and the propensity score ρ_k of §3.5, so it varies the ladder only through a scalar markup and posts higher average prices than the uniform control. Practical markup bounds (−5% to +20% of base price) are imposed per the Airline's guidelines. Models are calibrated on historical data from January 2025 and deployed without recalibration; the 4–8 month gap is mitigated by the slow-moving nature of the structural parameters, the Single Index (SI) refinement procedure (§EC.11), and a fallback mechanism for implausible recommendations. The analysis focuses on booking sessions to avoid strategic-behavior confounds in post-booking sessions (§EC.12).

Summary statistics. Table 2 presents summary statistics by treatment arm. The four arms are broadly comparable, supporting session-level randomization. Revenue and airfare are reported as a percentage of the control-arm mean.

Table 2 Summary statistics of main experiment by treatment arm

	FEA		SEG		PCP		Control	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Revenue (% of control)	104.91	128.30	106.54	131.43	101.02	139.58	100.00	116.79
Number of passengers	1.634	1.083	1.658	1.104	1.629	1.076	1.688	1.163
Trip days	3.989	10.127	3.858	8.637	4.377	13.498	3.743	8.600
Days to departure	48.146	52.613	49.347	52.434	49.016	53.177	50.874	55.320
Airfare (% of control)	95.17	87.93	96.08	92.18	96.09	83.42	100.00	96.23
N	33,937		18,377		3,408		3,419	

Notes: Each observation is identified by PNR, itinerary segment (OD), and segment departure date. Revenue and airfare are shown as a percentage of the control-arm mean.

[1] Two pilot studies on a single short-haul route preceded the main experiment. Pilot 1 compares the feature-based rule and the PCP rule against a uniform-pricing baseline, while Pilot 2 compares the feature-based and segment-based rules, together with the PCP rule, against a multi-armed bandit baseline. Full pilot details are reported in §EC.10.

6.2. Revenue Effects

Table 3 reports the estimated treatment effects on baggage revenue per booking session. Under the most comprehensive specification, FEA raises revenue per session by 7.75% of the control mean, SEG by 7.91%, and PCP by 7.39%, approximately 7–8% in each case. All three rules significantly outperform the uniform-pricing control. The three rules are quantitatively close, consistent with the theoretical prediction that FEA captures most of the gains from the fully segmented policy in sparse-data markets.

Table 3 Estimated treatment effect on revenue

	FEA			SEG			PCP		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treatment	4.444*	5.644**	7.751***	6.533**	6.957***	7.912***	0.871	3.768	7.389**
	(2.524)	(2.484)	(2.301)	(2.677)	(2.615)	(2.392)	(3.742)	(3.672)	(3.318)
Number of passengers			55.371***			47.869***			49.190***
			(1.540)			(2.045)			(3.610)
Trip days			0.578***			0.615***			0.358**
			(0.124)			(0.208)			(0.157)
Days to departure			−0.029			0.000			−0.154
			(0.059)			(0.132)			(0.106)
Airfare			0.077***			0.051***			0.059***
			(0.007)			(0.011)			(0.018)
Controls	No	No	Yes	No	No	Yes	No	No	Yes
Feature FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
OD FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	37,356	37,356	37,356	21,796	21,796	21,796	6,827	6,827	6,827
PNRs	26,697	26,697	26,697	15,552	15,552	15,552	4,799	4,799	4,799
Adjusted R^2	0.080	0.118	0.341	0.093	0.137	0.310	0.082	0.120	0.305

Notes. This table presents the estimated treatment effect on revenue relative to uniform pricing (control). The outcome is revenue per session as a percentage of the control-arm mean, so coefficients and standard errors are in percentage points of that mean (rescaled from the estimates in the original currency). Controls include fixed effects of day of week of the trip, trip type (e.g., one way, round trip), flight type (connecting or not), fare class, payment currency, and aircraft type of the flight. Date FE is the fixed effects of logging date. Feature FEs include fixed effects of all features. Cluster-adjusted standard errors at PNR level are shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The revenue gains arise from simultaneous increases in conversion and tier migration: differentiated markups via (5) lower prices on elastic tiers (attracting purchases) while raising prices on inelastic tiers (extracting surplus).

Posted-price distributions. Figure 1 reports the cumulative distributions of average price across the five products, by treatment arm. The FEA and SEG CDFs lie above the control CDF over most of the range, while the PCP CDF lies below; FEA and SEG therefore first-order stochastically dominate PCP. Combined with the observed revenue gains, the lower posted prices under FEA and SEG indicate that the joint optimization of base prices and feature markups simultaneously improves seller revenue and reduces the average price paid. Under attention-aware pricing, markups are reallocated toward inelastic tiers rather than extracted uniformly from elastic ones. The tier-by-tier mean ordering (average propensity prices above average RI prices) is established theoretically by Proposition 2; the stronger near-FOSD ordering visible in Figure 1 is an empirical pattern read directly from the cumulative distributions.

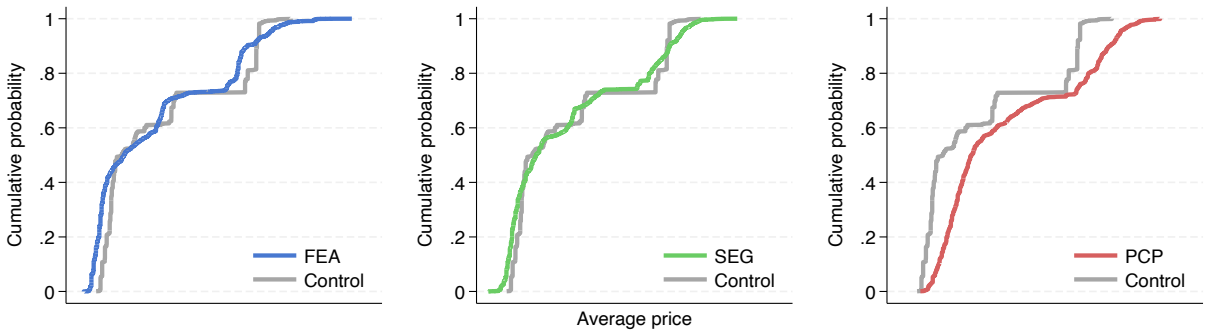


Figure 1 Cumulative distributions of average price by treatment arm. Both FEA and SEG first-order stochastically dominate the propensity-based (PCP) rule (cf. §3.5): lower average prices with higher revenue.

Treatment effects are positive across all but two regions (the smallest samples) and are strongest in medium-haul ODs. Regional and haul-type patterns are summarized in §EC.13.

6.3. Quantity Effects

The motivating evidence in Section 2 documents that consumers attend to baggage tiers heterogeneously. If the pricing framework exploits this heterogeneity, it should raise not only revenue but also total baggage quantity by lowering prices on tiers where attention, and hence demand elasticity, is greatest. Table EC.1 tests this prediction directly. FEA raises total quantity purchased by 0.0538 units per session relative to the control, SEG by 0.0348 units, and PCP by 0.0525 units, indicating that the revenue gain reflects expanded market participation rather than redistribution of surplus across tiers.

FEA versus SEG by sample size. FEA tends to outperform SEG on origin–destination pairs with fewer than approximately 22,000 observations; above this threshold the two rules perform similarly. The pattern is consistent with Theorem 4: FEA's variance advantage dominates in data-sparse markets, while SEG's bias advantage dominates when data are abundant.

6.4. Managerial Insights from Calibrated Parameters

An auxiliary parameter-fitting study on a distinct, time-limited historical extract of the Airline's directional ODs (30 of which stabilized within the convergence threshold) exposes a clear managerial pattern. The median attention weight α rises monotonically with route length, from 23.6 on very short-haul routes to 170.3 on long-haul routes, implying larger markups on longer routes via (5). This gradient matches the RI reading that short-haul passengers, who are typically frequent and price-conscious, attend more to baggage options than long-haul passengers with committed luggage needs. Full calibration, the tier-spread patterns, and this auxiliary dataset's relation to the 48-OD field experiment are reported in §EC.14.

7. Conclusions

This paper studies data-driven pricing for a class of multi-product problems characterized by horizontally differentiated substitutes, heterogeneous product-level price sensitivity, and sparse segment-level data, with airline checked baggage as the running example. Randomized price variation rejects the homogeneous-substitution benchmark and supports a generalized rational-inattention specification with product-level attention heterogeneity, an instance of the Uniformly Posterior-Separable class of Caplin et al. (2022) that also admits an equivalent representation as a marginal-distribution model (Mishra et al. 2014). Around this demand specification, the framework assembles four components: an LP-based estimation pipeline that admits a three-term error decomposition (merging bias, approximation bias, statistical error); a semi-closed-form optimal price ladder; a quadratic revenue-loss bound linking calibration error to revenue loss; and three implementable pricing rules of increasing data appetite. Synthetic experiments show that the framework recovers MNL behavior under homogeneous substitution and delivers substantial gains when product-level heterogeneity is present. The field experiment covers 48 origin–destination pairs. The feature-based and segment-based rules each raise baggage revenue by 7–8% over uniform pricing ($p < 0.01$); the propensity-based rule of §5.3 (the PCP arm) is included as the personalized-pricing benchmark. The posted-price distribution under the proposed rules first-order stochastically dominates the PCP distribution, so average posted prices fall while revenue rises. The quantity results confirm that the effect reflects expanded market participation rather than surplus extraction.

The framework's scope is delimited in three respects. The model abstracts from bundling, capacity constraints, and dynamic intertemporal interactions; the empirical evidence is drawn from a single carrier and a single ancillary category; and the semi-closed-form price ladder relies on the within-segment normalization $\gamma_{ik} = \gamma_k$. Richer choice environments and sharper attention-versus-taste diagnostics are natural directions for future work. Beyond airline ancillaries, the modeling and estimation pipeline applies directly to other multi-product pricing problems with substitution heterogeneity and sparse contextual data, including retail product lines, subscription tiers, and add-on menus.

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E-companion to "Multi-Product Pricing under Rational Inattention: Evidence, Estimation, and Field Deployment from Airline Ancillaries"

EC.1. Proofs of Main Results and Auxiliary Lemmas

This section collects the proofs of all main-text theorems and lemmas. Auxiliary lemmas needed exclusively in the revenue-loss analysis are stated and proved in §EC.1.12.

Notation. Throughout this section, proofs are stated for a generic segment with parameters $(\boldsymbol{v}, \boldsymbol{\alpha}, \boldsymbol{c})$; the feature-group specialization of §4.3 is recovered by setting $v_i = v_i^{(h)} + \boldsymbol{\beta}^{(h)\top}\boldsymbol{x}_g$, $\alpha_i = \alpha_i^{(h)}$, and $c_i = c_i^{(h)}$. Subscript k denotes a fine segment where relevant; subscript g denotes a g-group.

EC.1.1. Proof of Lemma 1

Proof. In the horizontal case there is a single dual multiplier μ for the simplex constraint. The buyer's KKT condition for the no-purchase option reads $0 - \alpha_0(\ln s_0 + 1) - \mu = 0$ at an interior optimum, so $\mu = -\alpha_0(\ln(1-S)+1)$ where $S = \sum_{i\in[I]} s_i$. For each product $i \in [I]$, the buyer's KKT condition $v_i - p_i - \alpha_i(\ln s_i + 1) - \mu = 0$ then gives

$$p_i \;=\; v_i - \alpha_i(\ln s_i + 1) + \alpha_0(\ln(1-S)+1) \;=\; w_i - \alpha_i \ln s_i + \alpha_0 \ln(1-S),$$

using $w_i = v_i - \alpha_i + \alpha_0$. Substituting into $\pi = \sum_{i\in[I]} s_i(p_i - c_i)$ yields the stated expression. □

EC.1.2. Proof of Proposition 1

Proof. Let $S := \sum_{i\in[I]} s_i$. The Hessian of the profit function in Lemma 1 is

$$\nabla^2_{\boldsymbol{ss}}\pi(\boldsymbol{s};\boldsymbol{\theta}) = -\operatorname{diag}\left(\frac{\alpha_1}{s_1}, \ldots, \frac{\alpha_I}{s_I}\right) - \frac{\alpha_0(2-S)}{(1-S)^2}\,\mathbf{1}\mathbf{1}^\top.$$

Since $\alpha_i > 0$, $s_i > 0$, and $S < 1$ on the interior of the feasible set, the diagonal term is negative definite and the rank-one term is negative semidefinite. Therefore $\nabla^2_{\boldsymbol{ss}}\pi(\boldsymbol{s};\boldsymbol{\theta})$ is negative definite on the interior of the feasible region, so $\pi(\boldsymbol{s};\boldsymbol{\theta})$ is strictly concave there. Extending π continuously to the closed feasible set via $0\ln 0 := 0$ preserves concavity (the closure of a strictly concave open-domain function is concave), so problem (4) is a convex optimization problem. □

EC.1.3. Proof of Theorem 1

Proof. We prove the claim for the slightly more general objective in which the intercepts are shifted by a common additive scalar $z \in \mathbb{R}$. Setting $z = 0$ recovers the statement of Theorem 1; setting $z = \boldsymbol{\beta}^{(h)\top}\boldsymbol{x}_g$ recovers the g-level feature-group version of §4.3. Write

$$S := \sum_{i=1}^{I} s_i, \qquad \pi(\boldsymbol{s}) = \textstyle\sum_{i\in[I]} s_i\big(w_i + z - \alpha_i \ln s_i + \alpha_0 \ln(1-S) - c_i\big).$$

By Proposition 1, π is strictly concave on the open interior $\{\boldsymbol{s} > 0 : S < 1\}$ and continuously extended to the closed feasible set via $0 \ln 0 := 0$. The extended objective is upper semi-continuous on the compact feasible set $\{\boldsymbol{s} \geq 0 : S \leq 1\}$, so by Weierstrass it admits a maximizer; uniqueness follows from strict concavity on the interior since the supremum cannot be attained on the boundary, as we show next. Moreover, the coupling term $\alpha_0 S \ln(1-S)$ tends to $-\infty$ as $S \uparrow 1$, so the optimum satisfies $S^\star < 1$. Also, for any $j \in [I]$, the derivative

$$\frac{\partial \pi}{\partial s_j} = w_j + z - \alpha_j(\ln s_j + 1) + \alpha_0 \ln(1-S) - \alpha_0 \frac{S}{1-S} - c_j$$

diverges to $+\infty$ as $s_j \downarrow 0$, so every optimal component satisfies $s_j^\star > 0$. Hence the first-order conditions are necessary and sufficient.

For each $j \in [I]$, the first-order condition is

$$0 = w_j + z - \alpha_j(\ln s_j^\star + 1) + \alpha_0 \ln(1-S^\star) - \alpha_0 \frac{S^\star}{1-S^\star} - c_j.$$

Rearranging gives

$$\ln s_j^\star = \frac{w_j + z + T^\star - c_j}{\alpha_j} - 1, \qquad s_j^\star = \exp\left(\frac{w_j + z + T^\star - c_j}{\alpha_j} - 1\right),$$

where

$$T^\star := \alpha_0 \ln(1-S^\star) - \alpha_0 \frac{S^\star}{1-S^\star}.$$

The term $T^\star$ is common across products and depends only on the aggregate share $S^\star$.

Recover prices from the KKT relation in Lemma 1:

$$p_i^\star = w_i + z - \alpha_i \ln s_i^\star + \alpha_0 \ln(1-S^\star).$$

Substituting the expression for $\ln s_i^\star$ yields

$$\begin{aligned} p_i^\star &= w_i + z - \alpha_i \left(\frac{w_i + z + T^\star - c_i}{\alpha_i} - 1\right) + \alpha_0 \ln(1-S^\star) \\ &= c_i + \alpha_i - T^\star + \alpha_0 \ln(1-S^\star) \\ &= c_i + \alpha_i + \alpha_0 \frac{S^\star}{1-S^\star}. \end{aligned}$$

This proves the semi-closed-form representation (5). In particular, all item-specific variation in optimal prices comes from $c_i + \alpha_i$; the intercepts w_i and the segment shifter z affect prices only through the common-markup term via $S^\star$.

Finally, for any $i, j \in [I]$,

$$p_i^\star - p_j^\star = (c_i - c_j) + (\alpha_i - \alpha_j).$$

Therefore the ordering of optimal prices coincides with the ordering of $c_i + \alpha_i$, and the stated monotonicity conclusions follow immediately. □

EC.1.4. Proof of Corollary 1

Proof. The proof of Theorem 1 already establishes the closed form $p_i^\star = c_i + \alpha_i + \alpha_0 S^\star(z)/(1 - S^\star(z))$ for the shifted objective (set $z \neq 0$); differentiating w.r.t. z gives the displayed derivative. Strict positivity of $\partial S^\star / \partial z$ follows from Lemma EC.4 in §EC.5. The relative-gap identity is immediate since $\partial(p_i^\star - p_j^\star)/\partial z = 0$.

□

EC.1.5. Proof of Proposition 2

Proof. Fix a tier i. The hypothesis $\bar{p}_i \geq \max_k p_{i,k}^{\mathrm{RI}}$ gives $p_{i,k}^{\mathrm{RI}} \leq \bar{p}_i$ pointwise in k, hence $\mathbb{E}_k[p_{i,k}^{\mathrm{RI}}] \leq \bar{p}_i$ by monotonicity of expectation. With the calibration $\boldsymbol{p}_0 = \boldsymbol{p}^{\mathrm{U}}$, the propensity-based rule of §3.5 is affine in ρ_k:

$$p_{i,k}^{\mathrm{Prop}} \;=\; p_i^{\mathrm{U}} + (\bar{p}_i - p_i^{\mathrm{U}})\,\frac{\rho_k}{\rho_{\max}},$$

so linearity of expectation gives

$$\mathbb{E}_k[p_{i,k}^{\mathrm{Prop}}] \;=\; p_i^{\mathrm{U}} + (\bar{p}_i - p_i^{\mathrm{U}})\,\frac{\mathbb{E}_k[\rho_k]}{\rho_{\max}}.$$

Subtracting $\mathbb{E}_k[p_{i,k}^{\mathrm{RI}}]$ from both sides and dividing by $\bar{p}_i - p_i^{\mathrm{U}} > 0$, the inequality $\mathbb{E}_k[p_{i,k}^{\mathrm{RI}}] \leq \mathbb{E}_k[p_{i,k}^{\mathrm{Prop}}]$ is equivalent to

$$\frac{\bar{p}_i - \mathbb{E}_k[p_{i,k}^{\mathrm{RI}}]}{\bar{p}_i - p_i^{\mathrm{U}}} \;\geq\; 1 - \frac{\mathbb{E}_k[\rho_k]}{\rho_{\max}},$$

which is (M_i); the steps reverse, establishing the "if and only if" claim.

For the sufficient condition, suppose $\mathbb{E}_k[p_{i,k}^{\mathrm{RI}}] \leq p_i^{\mathrm{U}}$. Then $\bar{p}_i - \mathbb{E}_k[p_{i,k}^{\mathrm{RI}}] \geq \bar{p}_i - p_i^{\mathrm{U}}$, so

$$\frac{\bar{p}_i - \mathbb{E}_k[p_{i,k}^{\mathrm{RI}}]}{\bar{p}_i - p_i^{\mathrm{U}}} \;\geq\; 1 \;\geq\; 1 - \frac{\mathbb{E}_k[\rho_k]}{\rho_{\max}},$$

where the last inequality uses $\mathbb{E}_k[\rho_k] \geq 0$. Hence ($\mathrm{M}_i$) holds and the mean-dominance conclusion follows.

The argument is dimension-free: it uses only the per-tier ceiling $\bar{p}_i$ and the affine dependence of $p_{i,k}^{\mathrm{Prop}}$ on ρ_k, and applies tier-by-tier to the full multi-tier generalized-RI setting without modification. □

EC.1.6. Proof of Lemma 2

Proof. For a fixed option $i \in [I] \cup \{0\}$, let X_i denote the number of observations selecting option i among n independent sessions. Under Assumption 1, $X_i \sim \mathrm{Binomial}(n, s_i)$ and $\hat{s}_i = X_i/n$.

The event $|\ln \hat{s}_i - \ln s_i| > \epsilon$ is equivalent to

$$\hat{s}_i \notin [s_i e^{-\epsilon},\, s_i e^{\epsilon}].$$

Define the multiplicative deviations

$$\delta_+ := e^{\epsilon} - 1 > 0, \qquad \delta_- := 1 - e^{-\epsilon} > 0.$$

Then

$$\hat{s}_i > s_i e^{\epsilon} \iff X_i > n s_i (1+\delta_+), \qquad \hat{s}_i < s_i e^{-\epsilon} \iff X_i < n s_i (1-\delta_-).$$

Applying standard Chernoff bounds for binomial random variables gives

$$\mathbb{P}(\hat{s}_i > s_i e^{\epsilon}) = \mathbb{P}\big(X_i > n s_i (1+\delta_+)\big) \le \left[\frac{\exp(\delta_+)}{(1+\delta_+)^{1+\delta_+}}\right]^{n s_i},$$
$$\mathbb{P}(\hat{s}_i < s_i e^{-\epsilon}) = \mathbb{P}\big(X_i < n s_i (1-\delta_-)\big) \le \left[\frac{\exp(-\delta_-)}{(1-\delta_-)^{1-\delta_-}}\right]^{n s_i}.$$

Since $1+\delta_+ = e^{\epsilon}$ and $1-\delta_- = e^{-\epsilon}$, the corresponding exponents simplify to

$$\ln\left(\frac{\exp(\delta_+)}{(1+\delta_+)^{1+\delta_+}}\right) = \delta_+ - (1+\delta_+)\ln(1+\delta_+) = -\big[1+(\epsilon-1)e^{\epsilon}\big],$$
$$\ln\left(\frac{\exp(-\delta_-)}{(1-\delta_-)^{1-\delta_-}}\right) = -\delta_- - (1-\delta_-)\ln(1-\delta_-) = -\big[1-(1+\epsilon)e^{-\epsilon}\big].$$

Therefore,

$$\mathbb{P}(\hat{s}_i > s_i e^{\epsilon}) \le \exp\Big(-n s_i \big[1+(\epsilon-1)e^{\epsilon}\big]\Big),$$
$$\mathbb{P}(\hat{s}_i < s_i e^{-\epsilon}) \le \exp\Big(-n s_i \big[1-(1+\epsilon)e^{-\epsilon}\big]\Big).$$

Now apply a union bound over all $I+1$ options. Let $s_{\min} := \min_{i\in[I]\cup\{0\}} s_i$. Then

$$\begin{aligned}\mathbb{P}\left(\bigcup_{i\in[I]\cup\{0\}} \{|\ln\hat{s}_i - \ln s_i| > \epsilon\}\right) &\le \sum_{i\in[I]\cup\{0\}} \mathbb{P}(\hat{s}_i > s_i e^{\epsilon}) + \sum_{i\in[I]\cup\{0\}} \mathbb{P}(\hat{s}_i < s_i e^{-\epsilon}) \\ &\le (I+1)\exp\Big(-n s_{\min}\big[1+(\epsilon-1)e^{\epsilon}\big]\Big) \\ &\quad + (I+1)\exp\Big(-n s_{\min}\big[1-(1+\epsilon)e^{-\epsilon}\big]\Big).\end{aligned}$$

Because

$$1-(1+\epsilon)e^{-\epsilon} \le 1+(\epsilon-1)e^{\epsilon} \qquad (\epsilon>0),$$

the lower-tail constant is the smaller one and therefore gives the tighter sample-size requirement; using it as an upper bound for both the upper- and lower-tail sums, the total tail probability is at most $2(I+1)\exp\big(-n s_{\min}[1-(1+\epsilon)e^{-\epsilon}]\big)$. Requiring this $\le \eta$ (equivalently, each of the two $(I+1)$-term sums $\le \eta/2$) gives

$$(I+1)\exp\Big(-n s_{\min}\big[1-(1+\epsilon)e^{-\epsilon}\big]\Big) \le \frac{\eta}{2}.$$

Solving for n yields

$$n \ge \frac{\ln\big(2(I+1)/\eta\big)}{\big[1-(1+\epsilon)e^{-\epsilon}\big]s_{\min}}.$$

Under this condition,

$$\mathbb{P}(|\ln\hat{s}_i - \ln s_i| \le \epsilon \text{ for all } i \in [I]\cup\{0\}) \ge 1-\eta.$$

EC.1.7. Proof of Theorem 2

Proof. Let

$$\Delta\boldsymbol{\theta}_k := \hat{\boldsymbol{\theta}}_k - \boldsymbol{\theta}_k^\star.$$

The proof has three steps: first relate $\boldsymbol{A}_k\Delta\boldsymbol{\theta}_k$ to the empirical optimization error, then control the perturbation terms caused by replacing $\boldsymbol{A}_k$ with $\hat{\boldsymbol{A}}_k$, and finally use the ℓ_1-minimum gain of $\boldsymbol{A}_k$ to convert this residual bound into a parameter bound.

Step 1: Start from optimality of the empirical LP. Because $\hat{\boldsymbol{\theta}}_k$ minimizes the empirical ℓ_1 loss over Θ,

$$\|\hat{\boldsymbol{A}}_k\hat{\boldsymbol{\theta}}_k - \boldsymbol{\nu}\|_1 \;\le\; \|\hat{\boldsymbol{A}}_k\boldsymbol{\theta}_k^\star - \boldsymbol{\nu}\|_1.$$

Using $\hat{\boldsymbol{A}}_k = \boldsymbol{A}_k + E_k$ and $\boldsymbol{\nu} = \boldsymbol{A}_k\boldsymbol{\theta}_k^\star + \boldsymbol{\rho}_k$, we obtain

$$\begin{aligned}\hat{\boldsymbol{A}}_k\hat{\boldsymbol{\theta}}_k - \boldsymbol{\nu} &= (\boldsymbol{A}_k + E_k)\hat{\boldsymbol{\theta}}_k - (\boldsymbol{A}_k\boldsymbol{\theta}_k^\star + \boldsymbol{\rho}_k)\\ &= \boldsymbol{A}_k(\hat{\boldsymbol{\theta}}_k - \boldsymbol{\theta}_k^\star) + E_k\hat{\boldsymbol{\theta}}_k - \boldsymbol{\rho}_k\\ &= \boldsymbol{A}_k\Delta\boldsymbol{\theta}_k + E_k\hat{\boldsymbol{\theta}}_k - \boldsymbol{\rho}_k. \qquad \text{(EC.1)}\end{aligned}$$

Rearranging (EC.1) gives

$$\boldsymbol{A}_k\Delta\boldsymbol{\theta}_k = (\hat{\boldsymbol{A}}_k\hat{\boldsymbol{\theta}}_k - \boldsymbol{\nu}) - E_k\hat{\boldsymbol{\theta}}_k + \boldsymbol{\rho}_k,$$

and therefore, by the triangle inequality,

$$\|\boldsymbol{A}_k\Delta\boldsymbol{\theta}_k\|_1 \le \|\hat{\boldsymbol{A}}_k\hat{\boldsymbol{\theta}}_k - \boldsymbol{\nu}\|_1 + \|E_k\hat{\boldsymbol{\theta}}_k\|_1 + \|\boldsymbol{\rho}_k\|_1. \qquad \text{(EC.2)}$$

Step 2: Bound the empirical residual and perturbation terms. By optimality of $\hat{\boldsymbol{\theta}}_k$,

$$\begin{aligned}\|\hat{\boldsymbol{A}}_k\hat{\boldsymbol{\theta}}_k - \boldsymbol{\nu}\|_1 &\le \|\hat{\boldsymbol{A}}_k\boldsymbol{\theta}_k^\star - \boldsymbol{\nu}\|_1\\ &= \|(\boldsymbol{A}_k + E_k)\boldsymbol{\theta}_k^\star - (\boldsymbol{A}_k\boldsymbol{\theta}_k^\star + \boldsymbol{\rho}_k)\|_1\\ &= \|E_k\boldsymbol{\theta}_k^\star - \boldsymbol{\rho}_k\|_1\\ &\le \|E_k\boldsymbol{\theta}_k^\star\|_1 + \|\boldsymbol{\rho}_k\|_1. \qquad \text{(EC.3)}\end{aligned}$$

Since only the $\boldsymbol{\alpha}$-block columns of E_k are nonzero, for every $\boldsymbol{\theta}_k \in \mathbb{R}^N$ of the form (6) we have

$$E_k\boldsymbol{\theta}_k = E_k\,\iota_\alpha(\boldsymbol{\alpha}_k),$$

where $\iota_\alpha : \mathbb{R}^{I+1} \to \mathbb{R}^N$ is the zero-padded embedding that places $\boldsymbol{\alpha}_k$ in the $\boldsymbol{\alpha}$-block and zeros elsewhere. Hence

$$\|E_k\boldsymbol{\theta}_k^\star\|_1 = \|E_k\,\iota_\alpha(\boldsymbol{\alpha}_k^\star)\|_1 \le \|E_k\|_{1\to1}\,\|\boldsymbol{\alpha}_k^\star\|_1, \qquad \text{(EC.4)}$$

and similarly

$$\|E_k\hat{\boldsymbol{\theta}}_k\|_1 = \|E_k\,\iota_\alpha(\hat{\boldsymbol{\alpha}}_k)\|_1 \le \|E_k\|_{1\to1}\,\|\hat{\boldsymbol{\alpha}}_k\|_1. \qquad \text{(EC.5)}$$

Substituting (EC.3)–(EC.5) into (EC.2) yields

$$\|\boldsymbol{A}_k\Delta\boldsymbol{\theta}_k\|_1 \le \|E_k\|_{1\to1}\,\|\boldsymbol{\alpha}_k^\star\|_1 + \|E_k\|_{1\to1}\,\|\hat{\boldsymbol{\alpha}}_k\|_1 + 2\|\boldsymbol{\rho}_k\|_1. \qquad \text{(EC.6)}$$

Step 3: Convert the residual bound into a parameter bound. By definition of the ℓ_1-minimum gain,

$$\|\boldsymbol{A}_k d\|_1 \geq \kappa_1(\boldsymbol{A}_k)\|d\|_1 \qquad \text{for every } d \in \mathbb{R}^N.$$

The inequality applies to $d = \Delta\boldsymbol{\theta}_k = \hat{\boldsymbol{\theta}}_k - \boldsymbol{\theta}_k^\star$ even though both $\hat{\boldsymbol{\theta}}_k, \boldsymbol{\theta}_k^\star \in \Theta$ (with $\boldsymbol{\alpha} \geq 0$, $\boldsymbol{\mu} \geq 0$): the difference $\Delta\boldsymbol{\theta}_k \in \mathbb{R}^N$ is unrestricted, and κ_1 is defined over all of $\mathbb{R}^N$. Applying this to $d = \Delta\boldsymbol{\theta}_k$ and combining with (EC.6) gives

$$\kappa_1(\boldsymbol{A}_k)\|\Delta\boldsymbol{\theta}_k\|_1 \leq \|E_k\|_{1\to 1}\,\|\boldsymbol{\alpha}_k^\star\|_1 + \|E_k\|_{1\to 1}\,\|\hat{\boldsymbol{\alpha}}_k\|_1 + 2\|\boldsymbol{\rho}_k\|_1. \tag{EC.7}$$

Next,

$$\|\hat{\boldsymbol{\alpha}}_k\|_1 \leq \|\boldsymbol{\alpha}_k^\star\|_1 + \|\hat{\boldsymbol{\alpha}}_k - \boldsymbol{\alpha}_k^\star\|_1 \leq \|\boldsymbol{\alpha}_k^\star\|_1 + \|\Delta\boldsymbol{\theta}_k\|_1,$$

because the $\boldsymbol{\alpha}$-block is a subvector of $\boldsymbol{\theta}_k$. Plugging this into (EC.7) yields

$$\kappa_1(\boldsymbol{A}_k)\|\Delta\boldsymbol{\theta}_k\|_1 \leq 2\|E_k\|_{1\to 1}\,\|\boldsymbol{\alpha}_k^\star\|_1 + \|E_k\|_{1\to 1}\,\|\Delta\boldsymbol{\theta}_k\|_1 + 2\|\boldsymbol{\rho}_k\|_1.$$

Rearranging,

$$\big(\kappa_1(\boldsymbol{A}_k) - \|E_k\|_{1\to 1}\big)\|\Delta\boldsymbol{\theta}_k\|_1 \leq 2\big(\|E_k\|_{1\to 1}\,\|\boldsymbol{\alpha}_k^\star\|_1 + \|\boldsymbol{\rho}_k\|_1\big).$$

Assumption 2 implies $\kappa_1(\boldsymbol{A}_k) - \|E_k\|_{1\to 1} > 0$, so dividing both sides gives

$$\|\hat{\boldsymbol{\theta}}_k - \boldsymbol{\theta}_k^\star\|_1 = \|\Delta\boldsymbol{\theta}_k\|_1 \leq \frac{2\big(\|E_k\|_{1\to 1}\,\|\boldsymbol{\alpha}_k^\star\|_1 + \|\boldsymbol{\rho}_k\|_1\big)}{\kappa_1(\boldsymbol{A}_k) - \|E_k\|_{1\to 1}} = \mathcal{E}_k.$$

Explicit bound under share error ϵ. It remains to show that $\|E_k\|_{1\to 1} \leq L\epsilon$. Recall that $\|E\|_{1\to 1}$ is the maximum absolute column sum. Only the $\boldsymbol{\alpha}$-columns of E_k are perturbed. For each product $i \in [I] \cup \{0\}$ and price level $l \in [L]$, the corresponding perturbation enters through the diagonal matrix $\hat{D}_{kl} - D_{kl}$ and equals

$$\big|[\hat{D}_{kl}]_{ii} - [D_{kl}]_{ii}\big| = \big|\ln \hat{s}_{ikl} - \ln s_{ikl}\big| \leq \epsilon.$$

Thus each α_i-column has exactly one perturbed entry per price level and therefore at most L perturbed entries, each bounded by ϵ in absolute value. It follows that every perturbed column sum is at most $L\epsilon$, so

$$\|E_k\|_{1\to 1} \leq L\epsilon.$$

Substituting this into the preceding display yields the explicit bound (10).

EC.1.8. Proof of Lemma 3

Proof. For any fine segment $k \in g$, the noiseless fine-segment calibration satisfies $\boldsymbol{A}_k\boldsymbol{\theta}_k^\star = \boldsymbol{\nu}$, and the group noiseless residual is $\boldsymbol{\rho}_g = \boldsymbol{\nu} - \boldsymbol{A}_g\boldsymbol{\theta}_g^\star$. Hence

$$\boldsymbol{A}_g(\boldsymbol{\theta}_g^\star - \boldsymbol{\theta}_k^\star) = (\boldsymbol{\nu} - \boldsymbol{\rho}_g) - \boldsymbol{A}_g\,\boldsymbol{\theta}_k^\star = \boldsymbol{A}_k\boldsymbol{\theta}_k^\star - \boldsymbol{\rho}_g - \boldsymbol{A}_g\,\boldsymbol{\theta}_k^\star = -(\boldsymbol{A}_g - \boldsymbol{A}_k)\boldsymbol{\theta}_k^\star - \boldsymbol{\rho}_g.$$

Applying the ℓ_1-minimum gain:

$$\kappa_{1,g}\,\|\boldsymbol{\theta}_g^\star - \boldsymbol{\theta}_k^\star\|_1 \leq \|\boldsymbol{A}_g(\boldsymbol{\theta}_g^\star - \boldsymbol{\theta}_k^\star)\|_1 = \|(\boldsymbol{A}_g - \boldsymbol{A}_k)\boldsymbol{\theta}_k^\star + \boldsymbol{\rho}_g\|_1. \quad \square$$

EC.1.9. Proof of Theorem 3

Proof. By the triangle inequality in ℓ_1, $\|\hat{\boldsymbol{\theta}}_g - \boldsymbol{\theta}_k^\star\|_1 \le \|\boldsymbol{\theta}_g^\star - \boldsymbol{\theta}_k^\star\|_1 + \|\hat{\boldsymbol{\theta}}_g - \boldsymbol{\theta}_g^\star\|_1$. The first term is bounded by Lemma 3; the second by Theorem 2 applied to the group calibration with design matrix $\boldsymbol{A}_g$ and pooled sample size N_g. □

EC.1.10. Proof of Lemma 4

Proof. Apply the ℓ_1-minimum gain to the parameter difference:

$$\kappa_{1,g}\,\|\tilde{\boldsymbol{\theta}}_{g|h}^\star - \boldsymbol{\theta}_g^\star\|_1 \le \|\boldsymbol{A}_g(\tilde{\boldsymbol{\theta}}_{g|h}^\star - \boldsymbol{\theta}_g^\star)\|_1 = \|\boldsymbol{\rho}_g - \tilde{\boldsymbol{\rho}}_{g|h}\|_1,$$

where the equality uses $\boldsymbol{A}_g\tilde{\boldsymbol{\theta}}_{g|h}^\star = \boldsymbol{\nu} - \tilde{\boldsymbol{\rho}}_{g|h}$ and $\boldsymbol{A}_g\boldsymbol{\theta}_g^\star = \boldsymbol{\nu} - \boldsymbol{\rho}_g$. □

EC.1.11. Proof of Theorem 4

Proof. By the triangle inequality in ℓ_1, $\|\hat{\boldsymbol{\theta}}_{g|h} - \boldsymbol{\theta}_k^\star\|_1 \le \|\boldsymbol{\theta}_g^\star - \boldsymbol{\theta}_k^\star\|_1 + \|\tilde{\boldsymbol{\theta}}_{g|h}^\star - \boldsymbol{\theta}_g^\star\|_1 + \|\hat{\boldsymbol{\theta}}_{g|h} - \tilde{\boldsymbol{\theta}}_{g|h}^\star\|_1$. The first term is bounded by Lemma 3; the second by Lemma 4; for the third, write $\hat{\boldsymbol{\theta}}_{g|h} - \tilde{\boldsymbol{\theta}}_{g|h}^\star = L_g(\hat{\boldsymbol{\theta}}_h^{\mathrm{lin}} - \tilde{\boldsymbol{\theta}}_h^{\mathrm{lin},\star})$ where L_g is the linear induction map defined after (12). By the operator-norm inequality, $\|\hat{\boldsymbol{\theta}}_{g|h} - \tilde{\boldsymbol{\theta}}_{g|h}^\star\|_1 \le \|L_g\|_{1\to1}\|\hat{\boldsymbol{\theta}}_h^{\mathrm{lin}} - \tilde{\boldsymbol{\theta}}_h^{\mathrm{lin},\star}\|_1 \le \eta_{g|h}\,\mathcal{E}_{g|h}$, where the last step uses Theorem 2 applied to the h-level calibration with design matrix $\boldsymbol{A}_h^{\mathrm{lin}}$ and pooled sample size N_h. The operator-norm bound $\|L_g\|_{1\to1} \le \max(1, I\|\boldsymbol{x}_g\|_\infty)$ is established in (13). □

EC.1.12. Revenue-Loss Analysis

This section proves the supporting lemmas underlying Theorem 5 and then the theorem itself. Assumptions 3–5 are stated in the main text (Section 5).

Recall the gradient and Hessian in $\boldsymbol{s}$:

$$\begin{aligned}\nabla_{\boldsymbol{s}}\pi(\boldsymbol{s};\boldsymbol{\theta}) &= \boldsymbol{w} + \boldsymbol{\beta}^\top\boldsymbol{x}\cdot\mathbf{1} - \boldsymbol{\alpha}\odot(\ln\boldsymbol{s}+\mathbf{1}) + \alpha_0\ln(1-S)\,\mathbf{1} - \alpha_0\tfrac{S}{1-S}\,\mathbf{1} - c,\\ \nabla_{ss}^2\pi(s;\boldsymbol{\theta}) &= -\mathrm{diag}\Big(\tfrac{\alpha_i}{s_i}\Big) - \Big(\tfrac{\alpha_0}{1-S} + \tfrac{\alpha_0}{(1-S)^2}\Big)\mathbf{1}\mathbf{1}^\top.\end{aligned} \tag{EC.8}$$

LEMMA EC.1 (**Pointwise interiority and continuity**). *Under Assumptions 3 and 4, for each fixed $\boldsymbol{\theta}\in\boldsymbol{\Theta}$, the objective $\pi(\cdot;\boldsymbol{\theta})$ has a unique maximizer $\boldsymbol{s}^\star(\boldsymbol{\theta})$ over Ω, this maximizer is interior, and $\boldsymbol{s}^\star(\boldsymbol{\theta})$ depends continuously on $\boldsymbol{\theta}$.*

Proof. First, note that the Hessian (EC.8) is negative definite for any $\boldsymbol{s}\in\Omega$ when $\alpha_i > 0$ (diagonal term strictly negative) and $\alpha_0 \ge 0$ (rank-one negative semidefinite coupling). Hence $\pi(\cdot;\boldsymbol{\theta})$ is strictly concave and admits a unique maximizer.

Next, consider $S \uparrow 1$ with $s \in \Omega$; then $\alpha_0 S \ln(1-S) \to -\infty$. Thus the objective cannot be maximized at S arbitrarily close to 1, so for each fixed $\boldsymbol{\theta}$ the maximizer satisfies $1 - S^\star(\boldsymbol{\theta}) > 0$. Furthermore, for any coordinate i, as $s_i \downarrow 0$ (with the other coordinates fixed to keep $S < 1$), the marginal derivative

$$\frac{\partial \pi}{\partial s_i} = w_i + \boldsymbol{\beta}^\top \boldsymbol{x} - \alpha_i(\ln s_i + 1) + \alpha_0 \ln(1-S) - \alpha_0 \tfrac{S}{1-S} - c_i$$

satisfies $\partial\pi/\partial s_i \to +\infty$ because $-\alpha_i \ln s_i \to +\infty$, while all other terms are bounded. Hence the objective strictly increases away from 0, and by strict concavity each component of the unique maximizer is strictly positive.

Finally, continuity follows from Berge's maximum theorem because π is continuous in $(\boldsymbol{s}, \boldsymbol{\theta})$ and the feasible set $\Omega = \{\boldsymbol{s} \geq 0 : \sum_i s_i \leq 1\}$ is compact and independent of $\boldsymbol{\theta}$; uniqueness then implies continuity of the optimizer. □

Assumption 5 upgrades the pointwise interiority of Lemma EC.1 to a uniform statement over the parameter set used in the robustness theorem. Under that assumption we obtain uniform curvature and smoothness on the compact parameter space.

LEMMA EC.2 **(Curvature and smoothness constants).** *Under Assumptions 3–5, there exist constants $\mu > 0$ and $L_\pi < \infty$ such that for all $\boldsymbol{\theta} \in \boldsymbol{\Theta}$,*

$$\mu I \preceq -\nabla^2_{ss} \pi\big(s^\star(\boldsymbol{\theta}); \boldsymbol{\theta}\big) \preceq L_\pi I.$$

Explicit choices are

$$\mu := \frac{\underline{\alpha}}{\overline{s}}, \qquad L_\pi := \frac{\overline{\alpha}}{\underline{s}} + I\,\alpha_0 \left(\frac{1}{\delta} + \frac{1}{\delta^2}\right),$$

where $\underline{s}$ and δ are from Assumption 5, $\overline{s} := 1 - \delta$, and $\overline{\alpha} := \max_{i \in [I]} \alpha_i$ (bounded on $\boldsymbol{\theta}$ by Assumption 3).

The above result shows that $\pi(\cdot; \boldsymbol{\theta})$ is μ-strongly concave and L_π-smooth in $\boldsymbol{s}$ at $\boldsymbol{s}^\star(\boldsymbol{\theta})$

Proof. Evaluate (EC.8) at $s^\star(\boldsymbol{\theta})$. By Assumption 5, $s_i^\star \in [\underline{s}, \overline{s}]$ and $1 - S^\star \geq \delta$. For any vector $\boldsymbol{v}$,

$$-\boldsymbol{v}^\top \nabla^2_{ss} \pi\, \boldsymbol{v} = \sum_i \frac{\alpha_i}{s_i^\star} v_i^2 + \left(\frac{\alpha_0}{1-S^\star} + \frac{\alpha_0}{(1-S^\star)^2}\right)\left(\sum_i v_i\right)^2.$$

We obtain the lower bound by keeping only the diagonal part and using $\alpha_i \geq \underline{\alpha}$, $s_i^\star \leq \overline{s}$:

$$-\boldsymbol{v}^\top \nabla^2_{ss} \pi \boldsymbol{v} \;\geq\; \frac{\underline{\alpha}}{\overline{s}} \|\boldsymbol{v}\|_2^2.$$

Similarly, we obtain the upper bound using $\alpha_i \leq \overline{\alpha}$, $s_i^\star \geq \underline{s}$, and $(\sum_i v_i)^2 \leq I \|\boldsymbol{v}\|_2^2$:

$$-\boldsymbol{v}^\top \nabla^2_{ss} \pi \boldsymbol{v} \;\leq\; \frac{\overline{\alpha}}{\underline{s}} \|\boldsymbol{v}\|_2^2 + \alpha_0 \left(\frac{1}{\delta} + \frac{1}{\delta^2}\right) I \|\boldsymbol{v}\|_2^2.$$

This yields the stated bound on L_π. □

LEMMA EC.3 (**Mixed-norm Jacobian bound**). *Under Assumptions 3–5, adopt the structural parameterization $\boldsymbol{\theta} = (\boldsymbol{w}, \boldsymbol{\alpha}, \boldsymbol{c})$ in which π is written via Lemma 1. The Jacobian $J(\boldsymbol{s}; \boldsymbol{\theta}) := \nabla^2_{\boldsymbol{s}\boldsymbol{\theta}} \pi(\boldsymbol{s}; \boldsymbol{\theta})$ satisfies $\|J\|_2 \le G$ and $\|J\|_{1\to 2} \le G_1$ for all $\boldsymbol{s}$ with $s_i \ge \underline{s}$ and $1 - S \ge \delta$, where*

$$G \le \sqrt{2I + I\,(\ln \underline{s} + 1)^2 + I\left(|\ln \delta| + \tfrac{1-\delta}{\delta}\right)^2}, \quad G_1 = \max\left(1,\ |\ln \underline{s} + 1|,\ \left|\ln \delta - \tfrac{1-\delta}{\delta}\right|\sqrt{I}\right),$$

and $\|J\|_{1\to 2} := \max_j \|J_{\cdot,j}\|_2$ denotes the maximum column ℓ_2 norm.

Proof. Treating $\boldsymbol{w}$, $\boldsymbol{\alpha}$ (including α_0), and $\boldsymbol{c}$ as independent under the structural parameterization, differentiate $\nabla_{\boldsymbol{s}} \pi$ componentwise (cf. Lemma 1). The columns of J are: w_i-columns are e_i (unit vectors, $\|e_i\|_2 = 1$); α_i-columns are $-(\ln s_i + 1)e_i$ (norm $|\ln s_i + 1| \le |\ln \underline{s} + 1|$); the α_0-column is $(\ln(1-S) - \frac{S}{1-S})\,\mathbf{1}$ (norm $|\ln \delta - \frac{1-\delta}{\delta}|\sqrt{I}$); c_i-columns are $-e_i$ (norm 1). The spectral norm bound G follows from $\|J\|_2 \le \|J\|_F = \sqrt{\sum_j \|J_{\cdot,j}\|_2^2}$, and $G_1 = \max_j \|J_{\cdot,j}\|_2$. □

REMARK EC.1 (ROLE OF THE $\boldsymbol{c}$-COLUMNS). The leading $2I$ in the Frobenius bound for G collects I from the $\boldsymbol{w}$-columns and I from the $\boldsymbol{c}$-columns (each a unit vector $\pm e_i$ of squared norm 1); since $\Delta \boldsymbol{c} = 0$ at deployment, the $\boldsymbol{c}$-contribution does not propagate to the final revenue-loss bound, but is retained here so that the formula is consistent with the structural parameterization $(\boldsymbol{w}, \boldsymbol{\alpha}, \boldsymbol{c})$. The G_1 bound is unchanged because each $\boldsymbol{c}$-column has ℓ_2 norm 1, already captured by the leading "1" in the maximum.

Proof of Theorem 5

Proof. We first prove

$$\|\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}}) - \boldsymbol{s}^\star(\boldsymbol{\theta}^\star)\|_2 \ \le\ \frac{G_1}{\mu}\,\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^\star\|_1. \tag{EC.9}$$

By Lemma EC.2 the map $\boldsymbol{s} \mapsto \pi(\boldsymbol{s}; \boldsymbol{\theta})$ is μ-strongly concave and L_π-smooth on Ω, i.e., for all $\boldsymbol{s}, \boldsymbol{t} \in \Omega$,

$$\pi(\boldsymbol{t}; \boldsymbol{\theta}) \le \pi(\boldsymbol{s}; \boldsymbol{\theta}) + \nabla_{\boldsymbol{s}} \pi(\boldsymbol{s}; \boldsymbol{\theta})^\top (\boldsymbol{t} - \boldsymbol{s}) - \tfrac{\mu}{2}\|\boldsymbol{t} - \boldsymbol{s}\|_2^2,$$
$$\|\nabla_{\boldsymbol{s}} \pi(\boldsymbol{t}; \boldsymbol{\theta}) - \nabla_{\boldsymbol{s}} \pi(\boldsymbol{s}; \boldsymbol{\theta})\|_2 \le L_\pi \|\boldsymbol{t} - \boldsymbol{s}\|_2,$$

for each $\boldsymbol{\theta}$ in a neighborhood of $\boldsymbol{\theta}^\star$. Hence $F(\boldsymbol{s}, \boldsymbol{\theta}) := \nabla_{\boldsymbol{s}} \pi(\boldsymbol{s}; \boldsymbol{\theta})$ satisfies, for fixed $\boldsymbol{\theta}$ and all $\boldsymbol{s}, \boldsymbol{t} \in \Omega$,

$$\big(F(\boldsymbol{s}, \boldsymbol{\theta}) - F(\boldsymbol{t}, \boldsymbol{\theta})\big)^\top (\boldsymbol{s} - \boldsymbol{t}) \le -\mu\,\|\boldsymbol{s} - \boldsymbol{t}\|_2^2,$$

i.e., $-F$ is μ-strongly monotone.

By FOC, the optimal shares solve

$$F\big(\boldsymbol{s}^\star(\boldsymbol{\theta}), \boldsymbol{\theta}\big) = 0.$$

Define $\Delta\boldsymbol{\theta} := \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^\star$ and $\Delta\boldsymbol{s} := \boldsymbol{s}^\star(\hat{\boldsymbol{\theta}}) - \boldsymbol{s}^\star(\boldsymbol{\theta}^\star)$. Consider the identity

$$0 = F\big(\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}}), \hat{\boldsymbol{\theta}}\big) - F\big(\boldsymbol{s}^\star(\boldsymbol{\theta}^\star), \boldsymbol{\theta}^\star\big).$$

Add and subtract $F\big(\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}}),\boldsymbol{\theta}^\star\big)$ to obtain

$$0=\underbrace{F\big(\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}}),\hat{\boldsymbol{\theta}}\big)-F\big(\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}}),\boldsymbol{\theta}^\star\big)}_{\text{(I)}}+\underbrace{F\big(\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}}),\boldsymbol{\theta}^\star\big)-F\big(\boldsymbol{s}^\star(\boldsymbol{\theta}^\star),\boldsymbol{\theta}^\star\big)}_{\text{(II)}}. \tag{EC.10}$$

Term (I) is a pure $\boldsymbol{\theta}$-change at fixed $\boldsymbol{s}=\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}})$; by the vector mean-value theorem along the segment $\boldsymbol{\theta}_t:=\boldsymbol{\theta}^\star+t\Delta\boldsymbol{\theta}$,

$$\text{(I)}=\int_0^1 \nabla_{\boldsymbol{\theta}}F\big(\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}}),\boldsymbol{\theta}_t\big)\,dt\cdot\Delta\boldsymbol{\theta},$$

hence, by Lemma EC.3 and the fact that $\nabla_{\boldsymbol{\theta}}F(\boldsymbol{s},\boldsymbol{\theta})=J(\boldsymbol{s};\boldsymbol{\theta})$, and the mixed-norm bound $\|J\boldsymbol{x}\|_2\le\|J\|_{1\to2}\|\boldsymbol{x}\|_1$,

$$\|\text{(I)}\|_2\le\int_0^1\big\|J(\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}}),\boldsymbol{\theta}_t)\big\|_{1\to2}\,dt\,\|\Delta\boldsymbol{\theta}\|_1\le G_1\,\|\Delta\boldsymbol{\theta}\|_1. \tag{EC.11}$$

Term (II) is a pure $\boldsymbol{s}$-change at fixed $\boldsymbol{\theta}=\boldsymbol{\theta}^\star$. Since $-F$ is μ-strongly monotone,

$$\big(F(\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}}),\boldsymbol{\theta}^\star)-F(\boldsymbol{s}^\star(\boldsymbol{\theta}^\star),\boldsymbol{\theta}^\star)\big)^\top\times\big(\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}})-\boldsymbol{s}^\star(\boldsymbol{\theta}^\star)\big)\le-\mu\,\|\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}})-\boldsymbol{s}^\star(\boldsymbol{\theta}^\star)\|_2^2.$$

Taking absolute values and applying Cauchy–Schwarz,

$$\|\text{(II)}\|_2\,\|\Delta\boldsymbol{s}\|_2\ge\mu\,\|\Delta\boldsymbol{s}\|_2^2\quad\Rightarrow\quad\|\text{(II)}\|_2\ge\mu\,\|\Delta\boldsymbol{s}\|_2. \tag{EC.12}$$

Now take norms in (EC.10) and use the triangle inequality and (EC.11)–(EC.12):

$$\mu\,\|\Delta\boldsymbol{s}\|_2\;\le\;\|\text{(II)}\|_2\;=\;\|\text{(I)}\|_2\;\le\;G_1\,\|\Delta\boldsymbol{\theta}\|_1,$$

which yields (EC.9).

Next, by the L_π-smoothness of $\boldsymbol{s}\mapsto\pi(\boldsymbol{s};\boldsymbol{\theta}^\star)$ we have for any $\boldsymbol{t}$:

$$\pi(\boldsymbol{t};\boldsymbol{\theta}^\star)\ge\pi\big(\boldsymbol{s}^\star(\boldsymbol{\theta}^\star);\boldsymbol{\theta}^\star\big)+\nabla_{\boldsymbol{s}}\pi\big(\boldsymbol{s}^\star(\boldsymbol{\theta}^\star);\boldsymbol{\theta}^\star\big)^\top\big(\boldsymbol{t}-\boldsymbol{s}^\star(\boldsymbol{\theta}^\star)\big)-\tfrac{L_\pi}{2}\,\|\boldsymbol{t}-\boldsymbol{s}^\star(\boldsymbol{\theta}^\star)\|_2^2.$$

Since $\nabla_{\boldsymbol{s}}\pi(\boldsymbol{s}^\star(\boldsymbol{\theta}^\star);\boldsymbol{\theta}^\star)=0$, this reduces to

$$\pi(\boldsymbol{t};\boldsymbol{\theta}^\star)\ge\pi\big(\boldsymbol{s}^\star(\boldsymbol{\theta}^\star);\boldsymbol{\theta}^\star\big)-\tfrac{L_\pi}{2}\,\|\boldsymbol{t}-\boldsymbol{s}^\star(\boldsymbol{\theta}^\star)\|_2^2.$$

Apply this with $\boldsymbol{t}=\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}})$ to obtain

$$\pi\big(\boldsymbol{s}^\star(\boldsymbol{\theta}^\star);\boldsymbol{\theta}^\star\big)-\pi\big(\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}});\boldsymbol{\theta}^\star\big)\le\frac{L_\pi}{2}\,\|\boldsymbol{s}^\star(\hat{\boldsymbol{\theta}})-\boldsymbol{s}^\star(\boldsymbol{\theta}^\star)\|_2^2.$$

Combine with (EC.9) to get (14). □

Bridging to ℓ_1 calibration. Theorem 5 bounds the revenue loss directly in terms of $\|\Delta\boldsymbol{\theta}\|_1^2$ via the mixed-norm Jacobian $G_1 = \|J\|_{1\to 2}$, eliminating the need for the lossy norm inequality $\|\Delta\boldsymbol{\theta}\|_2 \le \|\Delta\boldsymbol{\theta}\|_1$. Here $\boldsymbol{\theta}$ is the structural parameterization $(\boldsymbol{w}, \boldsymbol{\alpha}, \boldsymbol{c})$ used in Lemma EC.3. Under ℓ_1 calibration, the upstream bounds (Theorems 3 and 4) deliver $\|\Delta\boldsymbol{\theta}^{\text{full}}\|_1 \le \mathcal{E}$ on the calibration parameterization $(\boldsymbol{v}, \boldsymbol{\alpha}, \boldsymbol{\mu})$; by the identity $\Delta w_i = \Delta v_i - \Delta\alpha_i + \Delta\alpha_0$ for $i \in [I]$, the linear map $T : (\boldsymbol{v}, \boldsymbol{\alpha}) \mapsto (\boldsymbol{w}, \boldsymbol{\alpha})$ has columns with ℓ_1-sums 1 (each v_i-column), 2 (each α_i-column, $i \in [I]$), and I+1 (the α_0-column, which enters $\boldsymbol{w}$ for every $i \in [I]$ and contributes to α_0 itself); hence $\|T\|_{1\to 1} = I$+1 and $\|\Delta\boldsymbol{\theta}\|_1 = \|\Delta(\boldsymbol{w}, \boldsymbol{\alpha}, \boldsymbol{c})\|_1 \le (I{+}1)\,\mathcal{E}$, so:

$$\text{Revenue loss} \;\le\; \frac{L_\pi\, G_1^2}{2\,\mu^2}\,(I{+}1)^2\,\big(\mathcal{E}\big)^2.$$

Since $G_1 = \max_j \|J_{\cdot,j}\|_2$ (the maximum column norm) is typically much smaller than $G \le \|J\|_F$ (the Frobenius bound on the spectral norm), the mixed-norm approach avoids a factor of up to $\sqrt{N}$ in the norm bridge and up to N in the quadratic bound. The residual $(I{+}1)^2$ factor reflects the dimension of the structural-to-calibration coordinate change and is independent of sample size n.

EC.2. Cross-Validation for Segmentation and Linearization

Theorems 3 and 4 clarify how the partition of booking features into segmentation features $\mathcal{J}_S$ and linear features $\mathcal{J}_L$ shapes the pooling–linearization trade-off in theory, but their bounds depend on unobservable oracle quantities and are not intended as a deployment-time selector. In practice we select the partition directly from data via a split-sample cross-validation (CV) procedure that requires no knowledge of true parameters and balances merging bias, approximation bias, and estimation error through held-out share-prediction accuracy.

1. **Split.** Randomly partition the observed choice data in each fine segment k into two equal halves: a *training* set and a *test* set.
2. **Evaluate each candidate configuration $(\mathcal{J}_S, \mathcal{J}_L)$.** For each h-group $h \in G_{\mathcal{J}_S \setminus \mathcal{J}_L}$:
   (a) For each $g \in G_h$, merge the training-set shares across all fine segments $k \in g$ to obtain $\bar{s}_{igl}^{\text{train}}$.
   (b) Calibrate: if $\mathcal{J}_L = \emptyset$, fit $\hat{\boldsymbol{\theta}}_g$ independently per group; otherwise, perform the h-level joint calibration to obtain $\hat{\boldsymbol{\theta}}_{g|h}$ for each $g \in G_h$.
   (c) Extract estimated parameters $(\hat{\boldsymbol{v}}_{g|h}, \hat{\boldsymbol{\alpha}}_{g|h})$ from the calibrated $\hat{\boldsymbol{\theta}}_{g|h}$. For each fine segment $k \in g$, use the model's equilibrium solver to compute predicted shares $\hat{\boldsymbol{s}}_{kl}(\hat{\boldsymbol{\theta}}_{g|h})$ at each price level l and compare with the *test*-set shares.

   The CV score of the configuration is the total share prediction error across all segments:

$$\text{CV}(\mathcal{J}_S, \mathcal{J}_L) = \sum_{g \in G_{\mathcal{J}_S}} \sum_{k \in g} \sum_{l \in [L]} \|\hat{\boldsymbol{s}}_{kl}(\hat{\boldsymbol{\theta}}_{g|h}) - \boldsymbol{s}_{kl}^{\text{test}}\|_1.$$

3. **Select.** Choose the configuration minimizing $\text{CV}(\mathcal{J}_S, \mathcal{J}_L)$.

The mechanism naturally trades off estimation error (penalized by noisy predicted shares under fine segmentation) against merging bias (penalized by systematic prediction error under coarse grouping). Since each feature can be excluded, segmented, or linearized, the total number of configurations is 3^J; when J is large, a greedy forward-selection heuristic reduces the search to $O(J^2)$ evaluations. The next section reports synthetic experiments confirming that the CV-selected configuration achieves 96%–99% of oracle revenue.

EC.3. Generalized RI and MDM Bridges

The demand model (1) admits two complementary microfoundations: (i) a generalized rational-inattention representation in which the information cost is the Bregman information generated by the weighted Shannon entropy, placing the model in the Uniformly Posterior-Separable class of Caplin et al. (2022); and (ii) a Marginal Distribution Model (MDM) representation (Mishra et al. 2014) with product-specific exponential noise marginals and a comonotone copula. Both representations recover the same conditional choice probabilities and the same calibrated weights α_{ik} as the APU form (2).

We fix a generic segment and suppress the subscript k throughout. Write $\boldsymbol{s} \in \Delta := \{\boldsymbol{s} \geq 0 : \sum_{i=0}^{I} s_i = 1\}$ for shares, $\boldsymbol{v} = (v_0, \ldots, v_I)$ for net utilities ($v_0 = 0$), and $\boldsymbol{\alpha} = (\alpha_0, \ldots, \alpha_I) \in \mathbb{R}^{I+1}_{++}$ for the per-tier weights. Following the convention introduced in §3.1, we use Φ (rather than Ω) to denote the weighted Shannon entropy in this section, reserving Ω exclusively for the feasible set $\{A\boldsymbol{s}^\top \leq \boldsymbol{b}\}$. Let

$$\Phi(\boldsymbol{s}) \;:=\; \sum_{i=0}^{I} \alpha_i \, s_i \ln s_i .$$

EC.3.1. Bregman Information Cost from the Weighted Entropy

The weighted entropy Φ is strictly convex on $\mathrm{int}(\Delta)$ ($\nabla^2\Phi(\boldsymbol{s}) = \mathrm{diag}(\alpha_i/s_i)$), smooth on $\mathrm{int}(\Delta)$, and satisfies the boundary-steepness condition $\|\nabla\Phi(\boldsymbol{s})\| \to \infty$ as $\boldsymbol{s} \to \partial\Delta$. Hence Φ is admissible in the sense of Hofbauer and Sandholm (2002) and is a valid generator for a Bregman divergence and a generalized entropy in the sense of Fosgerau et al. (2020).

Following Banerjee et al. (2005) and Fosgerau et al. (2020), define the Bregman divergence generated by Φ,

$$D_\Phi(\boldsymbol{p} \,\|\, \boldsymbol{q}) \;:=\; \Phi(\boldsymbol{p}) - \Phi(\boldsymbol{q}) - \nabla\Phi(\boldsymbol{q}) \cdot (\boldsymbol{p} - \boldsymbol{q}),$$

which, using $\nabla\Phi(\boldsymbol{q})_i = \alpha_i(\ln q_i + 1)$, admits the closed form

$$D_\Phi(\boldsymbol{p} \,\|\, \boldsymbol{q}) \;=\; \sum_{i=0}^{I} \alpha_i \, p_i \ln \tfrac{p_i}{q_i} \;-\; \sum_{i=0}^{I} \alpha_i \, (p_i - q_i). \tag{EC.13}$$

For $\alpha_i \equiv 1$ on the simplex the linear correction vanishes and D_Φ collapses to Shannon's KL divergence; for non-uniform α_i, D_Φ is a strictly weighted Bregman generalization of KL.

For a state-contingent choice strategy $\boldsymbol{p}(\cdot):\mathcal{V}\to\Delta$ with unconditional distribution $\boldsymbol{p}^0=\mathbb{E}_{\boldsymbol{V}}\,\boldsymbol{p}(\boldsymbol{V})$, the corresponding *Bregman information cost* (Fosgerau et al. 2020, eq. 14) is

$$\kappa_\Phi\big(\boldsymbol{p}(\cdot),\boldsymbol{p}^0\big)\;:=\;\mathbb{E}_{\boldsymbol{V}}\big[D_\Phi(\boldsymbol{p}(\boldsymbol{V})\,\|\,\boldsymbol{p}^0)\big]\;=\;\sum_{i=0}^{I}\alpha_i\,\mathbb{E}_{\boldsymbol{V}}\Big[p_i(\boldsymbol{V})\ln\tfrac{p_i(\boldsymbol{V})}{p_i^0}\Big]\;\geq 0,$$

where the second equality follows because the linear term in (EC.13) integrates to zero. This object is the Bregman information of Banerjee et al. (2005); it equals Shannon mutual information $I(\boldsymbol{V};A)$ only when $\alpha_i\equiv\lambda$ (and then $\kappa_\Phi=\lambda I(\boldsymbol{V};A)$), and is otherwise a per-tier-weighted generalization that lies in the Uniformly Posterior-Separable class of Caplin et al. (2022).

PROPOSITION EC.1 **(Generalized RI representation)**. *Fix any prior distribution $\boldsymbol{p}^0\in\operatorname{int}(\Delta)$ and consider the prior-anchored generalized rational-inattention problem*

$$\max_{\boldsymbol{p}(\cdot):\mathcal{V}\to\Delta}\;\mathbb{E}_{\boldsymbol{V}}\big[\boldsymbol{V}\cdot\boldsymbol{p}(\boldsymbol{V})\big]\;-\;\kappa_\Phi\big(\boldsymbol{p}(\cdot),\boldsymbol{p}^0\big).$$

The conditional choice probability $\boldsymbol{p}^\star(\boldsymbol{v})$ in state $\boldsymbol{v}$ coincides with the maximizer of the APU problem (2) evaluated at the Bregman-shifted utilities

$$\tilde v_i\;=\;v_i\;+\;\alpha_i(\ln p_i^0+1),\tag{EC.14}$$

and admits the closed form

$$p_i^\star(\boldsymbol{v})\;=\;p_i^0\,\exp\!\big((v_i-\mu(\boldsymbol{v}))/\alpha_i\big),\qquad \mu(\boldsymbol{v})\ \textit{solves}\ \sum_{i=0}^{I}p_i^0\,e^{(v_i-\mu)/\alpha_i}=1.\tag{EC.15}$$

When $\alpha_i\equiv\lambda$ this collapses to the prior-anchored multinomial logit of Matejka and McKay (2015), $p_i^\star\propto p_i^0\,e^{v_i/\lambda}$. The endogenous-prior version in which $\boldsymbol{p}^0$ is required to satisfy $\boldsymbol{p}^0=\mathbb{E}_{\boldsymbol{V}}\,\boldsymbol{p}^\star(\boldsymbol{V})$ is recovered as a fixed point of the per-state map above; this paper uses only the prior-anchored form, which suffices for the equivalence with (1) (take $\boldsymbol{p}^0$ uniform).

Proof. Expanding the Bregman divergence gives $D_\Phi(\boldsymbol{p}\,\|\,\boldsymbol{p}^0)=\Phi(\boldsymbol{p})-\langle\nabla\Phi(\boldsymbol{p}^0),\boldsymbol{p}\rangle+\big(\langle\nabla\Phi(\boldsymbol{p}^0),\boldsymbol{p}^0\rangle-\Phi(\boldsymbol{p}^0)\big)$, where the bracketed constant is $\boldsymbol{p}$-independent. Hence the per-state RI subproblem is, up to a constant,

$$\boldsymbol{v}\cdot\boldsymbol{p}\;-\;D_\Phi(\boldsymbol{p}\,\|\,\boldsymbol{p}^0)\;=\;\langle\,\boldsymbol{v}+\nabla\Phi(\boldsymbol{p}^0),\,\boldsymbol{p}\,\rangle\;-\;\Phi(\boldsymbol{p})\;+\;\text{const},$$

so its maximizer equals the APU maximizer at $\tilde v_i=v_i+\nabla\Phi(\boldsymbol{p}^0)_i=v_i+\alpha_i(\ln p_i^0+1)$, proving (EC.14). The first-order condition $\alpha_i\ln(p_i/p_i^0)=v_i-\mu$ yields the closed form (EC.15); the multiplier $\mu(\boldsymbol{v})$ is determined by $\sum_i p_i^\star=1$. The shift $\alpha_i(\ln p_i^0+1)$ is not of the form $c\cdot\mathbf{1}$ for non-uniform α_i and therefore does affect the optimizer. □

EC.3.2. MDM–Comonotone Equivalence

PROPOSITION EC.2 **(MDM–exponential representation)**. *Let* $F_i(x) = 1 - \exp(-x/\alpha_i)$ *for* $x \geq 0$. *The Marginal Distribution Model of Mishra et al. (2014) with deterministic utilities* $V_i = v_i - \alpha_i$ *and marginal CDFs* F_i *on the additive utility errors yields the same choice probabilities as the APU problem* (2).

Proof. The MDM choice probabilities maximize $\sum_i \left(V_i P_i + \int_{1-P_i}^{1} F_i^{-1}(t)\, dt\right)$ over the simplex (Mishra et al. 2014). With $F_i^{-1}(t) = -\alpha_i \ln(1-t)$, the substitution $u = 1 - t$ gives $\int_{1-P_i}^{1} F_i^{-1}(t)\, dt = \alpha_i(P_i - P_i \ln P_i)$, so the MDM objective becomes $\sum_i (V_i + \alpha_i)P_i - \sum_i \alpha_i P_i \ln P_i$. Setting $V_i = v_i - \alpha_i$ recovers (2). □

The corresponding random-utility joint distribution is the comonotone copula on the prescribed exponential marginals, a singular but valid joint; optimality of the comonotone arrangement follows from classical rearrangement results. Convexity of the pricing problem (Proposition 1) is established in §3.4 by direct Hessian calculation; the exponential density is log-concave, which fits the established pattern that log-concave noise densities are conducive to concave seller's problems under additive random-utility demand (see, e.g., Li and Huh 2011 for the MNL case).

REMARK EC.2 (SCOPE OF THE BRIDGE). The generalized-RI representation of Proposition EC.1 rests directly on the Bregman / UPS construction of Banerjee et al. (2005), Fosgerau et al. (2020), and Caplin et al. (2022), and does not require the regular-ARUM hypothesis of Fosgerau et al. (2020, Proposition 5). That hypothesis (absolute continuity with strictly positive density of the joint error distribution on $\mathbb{R}^{I+1}$) sits outside the comonotone joint of Proposition EC.2, whose support is one-dimensional. The two bridges are thus complementary: Proposition EC.1 supplies a rational-inattention microfoundation through the Bregman/UPS route, and Proposition EC.2 supplies a random-utility microfoundation through the comonotone construction. They agree on the calibrated α_i and on the choice probabilities induced from (1).

EC.4. Full Regression Tables for the Reduced-Form Evidence

This section reports (see Table EC.1) the complete regression tables underlying the reduced-form evidence summarized in §2. Hypotheses H1–H4 correspond to the four patterns documented there, and H5 tests the field experiment's quantity effects (§6.3). All controls and specifications are included; the main text reports only compact summary statistics.

EC.5. Structural Properties of Feature-Based Pricing

The feature-based pricing rule of §4.3 rests on two structural properties of the optimal solution: the optimal aggregate share $S^\star$ depends on the segment features only through a scalar shifter z_k, and the optimal price ladder shifts in parallel with z_k at a slope strictly below one. Let $z_k := \boldsymbol{\beta}^\top \boldsymbol{x}_k$ denote the (scalar) segment shift for segment k. For simplicity we drop the subscript k within the proofs and write z and $S^\star$. From Theorem 1 the feature $\boldsymbol{x}_k$ enters the optimal prices only indirectly via the aggregate share $S^\star(z_k)$; we therefore study the structural properties of $S^\star(z_k)$ in detail.

Table EC.1 Estimated treatment effect on total quantity purchased

	FEA	SEG	PCP
	(1)	(2)	(3)
Treatment	0.0538***	0.0348**	0.0525**
	(0.0169)	(0.0176)	(0.0230)
Number of passengers	0.2983***	0.2565***	0.2516***
	(0.0119)	(0.0142)	(0.0265)
Trip days	0.0024***	0.0028**	0.0010
	(0.0007)	(0.0012)	(0.0008)
Days to departure	0.0005	0.0005	−0.0008
	(0.0005)	(0.0007)	(0.0007)
Airfare	0.0004***	0.0003***	0.0003***
	(0.0000)	(0.0001)	(0.0001)
Controls	Yes	Yes	Yes
Feature FE	Yes	Yes	Yes
OD FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Observations	37,356	21,796	6,827
PNRs	26,697	15,552	4,799
Adjusted R^2	0.210	0.189	0.179

Notes: This table presents the estimated treatment effect on total quantity purchased. Controls include fixed effects of OD, fare class, and payment currency. Date FE is the fixed effects of booking date. Feature FEs include fixed effects of all six features. Cluster-adjusted standard errors at PNR level are shown in parentheses. ** $p < 0.05$, *** $p < 0.01$.

By the FOC of the profit maximization problem we have

$$\ln s_i^\star = \frac{w_i + z + T(S^\star) - c_i}{\alpha_i} - 1,$$

where $T(S) := \alpha_0 \ln(1-S) - \alpha_0 \frac{S}{1-S}$. Thus, we have $s_i(S,z) := \exp\left(\frac{w_i+z+T(S)-c_i}{\alpha_i} - 1\right) > 0$. Note that $S = \sum_i s_i$. Define

$$G(S,z) \;:=\; \sum_i s_i(S,z) - S = \sum_{i=1}^{I} \exp\left(\frac{w_i+z+T(S)-c_i}{\alpha_i} - 1\right) - S, S \in (0,1), \quad z \in \mathbb{R}.$$

Thus, solving for optimal $s_i^\star$ reduces to finding zeros of function $G(S,z)$; sufficiency of the FOC follows from strict concavity of $\pi(\cdot;\boldsymbol{\theta})$ on the open interior (Proposition 1). In the following, we show that the root is unique.

LEMMA EC.4 **(Existence and uniqueness of $S^\star$).** *For each z, there exists a unique $S^\star(z) \in (0,1)$ solving $G(S^\star(z), z) = 0$. Moreover, $S^\star(z)$ is continuously differentiable and strictly increasing in z:*

$$\frac{dS^\star}{dz} \;=\; -\frac{\partial_z G(S^\star, z)}{\partial_S G(S^\star, z)} \;>\; 0.$$

Proof. First, $T(S) = \alpha_0 \ln(1-S) - \alpha_0 \frac{S}{1-S}$ satisfies

$$T'(S) = -\alpha_0 \left(\frac{1}{1-S} + \frac{1}{(1-S)^2}\right) < 0, \text{for } S \in (0,1).$$

Hence

$$\partial_S G(S,z) = \sum_{i=1}^{I} s_i(S,z)\,\frac{T'(S)}{\alpha_i} - 1 \;<\; -1 \;<\; 0,$$

so $G(\cdot,z)$ is strictly decreasing on $(0,1)$. The boundary values are $\lim_{S\downarrow 0} G(S,z) = \sum_i \exp(\frac{w_i+z-c_i}{\alpha_i} - 1) > 0$ and $\lim_{S\uparrow 1} G(S,z) = -1$, since each $s_i(S,z) \to 0$ as $T(S) = \alpha_0 \ln(1-S) \to -\infty$ while $S \to 1$. Because G is continuous, strictly decreasing, and transitions from a positive value to -1, the intermediate value theorem yields a unique $S^\star(z) \in (0,1)$ with $G(S^\star(z),z) = 0$.

Next, $\partial_z G(S,z) = \sum_i \frac{1}{\alpha_i} s_i(S,z) > 0$ and $\partial_S G(S^\star,z) < 0$. By the implicit function theorem, $S^\star(z)$ is C^1 and

$$\frac{dS^\star}{dz} = -\frac{\partial_z G(S^\star,z)}{\partial_S G(S^\star,z)} > 0.$$

Moreover, since $T'(S)$ does not depend on i, the factor $|T'|$ cancels from the ratio:

$$\frac{dS^\star}{dz} = \frac{\sum_i s_i/\alpha_i}{|T'|\sum_i s_i/\alpha_i + 1} \le \frac{1}{|T'(S^\star)|} = \frac{(1-S^\star)^2}{\alpha_0\,(2-S^\star)}.$$

□

The quantitative bound on $dS^\star/dz$ immediately yields a bound on price sensitivity. Combined with Corollary EC.1:

$$\frac{\partial p_i^\star}{\partial z} = \frac{\alpha_0}{(1-S^\star)^2}\,\frac{dS^\star}{dz} \le \frac{1}{2-S^\star} \le 1.$$

That is, a unit increase in the segment shifter $z = \boldsymbol{\beta}^\top \boldsymbol{x}$ raises all prices by at most \$1.

Lemma EC.4 implies the comparative statics of optimal prices w.r.t. segment features summarized in the following

COROLLARY EC.1 **(Price sensitivity to feature shift)**. *Let $z_k = \beta^\top x_k$. For each i,*

$$\frac{\partial p_{i,k}^\star}{\partial z_k} = \alpha_0\,\frac{\partial}{\partial z_k}\Big(\frac{S_k^\star}{1-S_k^\star}\Big) = \alpha_0\,\frac{1}{(1-S_k^\star)^2}\,\frac{dS_k^\star}{dz_k} > 0,$$

where $\frac{dS_k^\star}{dz_k} > 0$ by Lemma EC.4. Thus, all optimal prices in segment k increase with the segment shifter z_k by the same amount (a common markup adjustment), while the relative ladder $p_i^\star - p_j^\star$ remains unchanged.

COROLLARY EC.2 **(Feature-coordinate gradient)**. *For any feature coordinate m,*

$$\frac{\partial p_{i,k}^\star}{\partial x_{k,m}} = \beta_m\,\frac{\partial p_{i,k}^\star}{\partial z_k} = \beta_m\,\alpha_0\,\frac{1}{(1-S_k^\star)^2}\,\frac{dS_k^\star}{dz_k}.$$

Hence if $\beta_m > 0$ (resp. < 0), increasing feature $x_{k,m}$ shifts the entire segment's price ladder up (resp. down) by a common amount.

Additional comparative statics follow directly: $\partial p_i^\star/\partial \alpha_i = 1$ (dollar-for-dollar markup), $S_{g_1}^\star < S_{g_2}^\star$ implies $p_{i,g_1}^\star < p_{i,g_2}^\star$ for all i, and products are approximately independently priced modulo the common markup.

Parallel shift pricing. Features affect prices only indirectly via $S_k^\star(\boldsymbol{x}_k)$: the ladder's *shape* (relative price gaps) is universal, while its *height* shifts with segment demand. This *parallel shift* structure, where the entire ladder moves together as $z_k = \boldsymbol{\beta}^\top \boldsymbol{x}_k$ varies, motivates the feature-based pricing policy. Introducing item-specific interactions ($\boldsymbol{\beta}_i^\top \boldsymbol{x}_k$) would break this structure.

EC.6. h-level Design Matrix Construction

This section gives the explicit block structure of the h-level design matrix $\hat{\boldsymbol{A}}_h^{\text{lin}}$ used in the feature-based calibration of §4.3.

h-level design matrix. Applying the linear model $v_{ig} = v_i^{(h)} + \boldsymbol{\beta}_{\mathcal{J}_L}^{(h)} \cdot \boldsymbol{x}_g$ for $i \in [I]$, the P-block of each g-level design matrix (7) splits into P (acting on $\boldsymbol{v}^{(h)}$) and $P\mathbf{1}_I \boldsymbol{x}_g^\top \in \mathbb{R}^{(I+1)\times|\mathcal{J}_L|}$ (acting on $\boldsymbol{\beta}_{\mathcal{J}_L}^{(h)}$), where $\mathbf{1}_I := (1,\dots,1)^\top \in \mathbb{R}^I$ and $P\mathbf{1}_I = (0,1,\dots,1)^\top$ reflects that the outside option receives no feature shift. The $\hat{D}_{gl}$ blocks now act on the common $\boldsymbol{\alpha}^{(h)}$, replacing the group-specific $\boldsymbol{\alpha}^{(g)}$ of the g-level calibration. The B_{gl} blocks remain group-specific, acting on the corresponding $\boldsymbol{\mu}^{(g)}$.

For each $g \in G_h$, define the shared-parameter block

$$\hat{\boldsymbol{C}}_g := \begin{pmatrix} P & P\mathbf{1}_I \boldsymbol{x}_g^\top & \hat{D}_{g1} \\ \vdots & \vdots & \vdots \\ P & P\mathbf{1}_I \boldsymbol{x}_g^\top & \hat{D}_{gL} \end{pmatrix} \in \mathbb{R}^{M\times(2I+1+|\mathcal{J}_L|)}$$

acting on $(\boldsymbol{v}^{(h)}, \boldsymbol{\beta}_{\mathcal{J}_L}^{(h)}, \boldsymbol{\alpha}^{(h)})$, and the group-specific $\boldsymbol{\mu}$-block $\hat{\boldsymbol{B}}_g \in \mathbb{R}^{M\times mL}$ with the same block-diagonal structure as in (7), acting on $\boldsymbol{\mu}^{(g)}$. The h-level design matrix is

$$\hat{\boldsymbol{A}}_h^{\text{lin}} = \begin{pmatrix} \hat{\boldsymbol{C}}_{g_1} & \hat{\boldsymbol{B}}_{g_1} & \mathbf{0} & \cdots & \mathbf{0} \\ \hat{\boldsymbol{C}}_{g_2} & \mathbf{0} & \hat{\boldsymbol{B}}_{g_2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \hat{\boldsymbol{C}}_{g_{|G_h|}} & \mathbf{0} & \mathbf{0} & \cdots & \hat{\boldsymbol{B}}_{g_{|G_h|}} \end{pmatrix}$$

with $\hat{\boldsymbol{A}}_h^{\text{lin}} \in \mathbb{R}^{|G_h|M\times(2I+1+|\mathcal{J}_L|+|G_h|mL)}$, and the calibration solves

$$\min_{\boldsymbol{\theta}_h^{\text{lin}}} \|\hat{\boldsymbol{A}}_h^{\text{lin}} \boldsymbol{\theta}_h^{\text{lin}} - \boldsymbol{\nu}_h\|_1, \qquad \boldsymbol{\nu}_h := (\boldsymbol{\nu}^\top,\dots,\boldsymbol{\nu}^\top)^\top.$$

The h-level parameterization introduces approximation error from two sources: the linear model on v_{ig} (measured by the residual $\delta_{ig}^{(h)} := v_{ig} - v_i^{(h)} - \boldsymbol{\beta}_{\mathcal{J}_L}^{(h)} \cdot \boldsymbol{x}_g$) and the sharing of $\boldsymbol{\alpha}^{(h)}$ across groups that may have heterogeneous $\boldsymbol{\alpha}^{(g)}$. The Lagrange multipliers $\boldsymbol{\mu}^{(g)}$ remain group-specific and introduce no additional approximation error. Compared with independent g-level calibrations, the shared parameter count drops from $(2I+1)\cdot|G_{\mathcal{J}_S}|$ to $(2I+1+|\mathcal{J}_L|)\cdot|G_{\mathcal{J}_S\setminus\mathcal{J}_L}|$, a significant reduction when $|G_{\mathcal{J}_S}|$ is large.

EC.7. Numerical Validation of the Cross-Validation Procedure

The split-sample CV procedure of §EC.2 is validated numerically as follows. The simulation uses $|\mathcal{J}_S| = 3$ features with mixed cardinality (one binary segmentation feature, one 4-level near-linear feature, one weak binary feature; $K = 16$ fine segments and 27 candidate configurations), $I = 3$ products, $L = 3$ price levels, and 100 replications per sample size $n \in \{200, 500, 1{,}000, 2{,}000, 5{,}000, 10{,}000, 20{,}000\}$. CV-selected configurations are compared against oracle, full-segmentation, and full-pooling baselines.

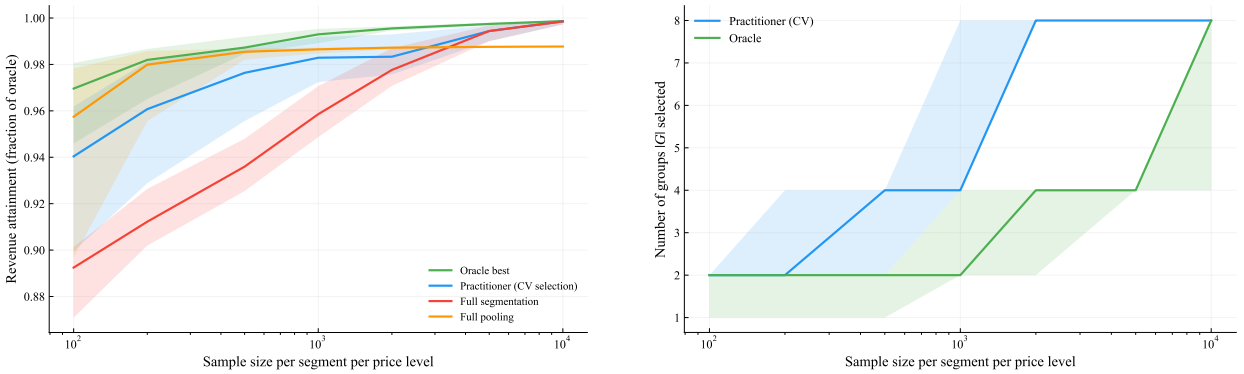


Figure EC.1 CV validation under moderate heterogeneity ($\sigma = 0.3$). *Left*: revenue attainment vs. sample size for CV (blue), oracle (green), full-segmentation (red), full-pooling (orange). *Right*: $|\mathcal{G}|$ selected by CV vs. oracle. Bands report IQR over 100 replications.

Moderate heterogeneity ($\sigma = 0.3$). The left panel of Figure EC.1 shows that CV achieves 96%–99% of oracle revenue across all sample sizes. The right panel confirms progressive refinement: CV selects coarser groupings at small n and refines as data grows, tracking the oracle trajectory.

High heterogeneity ($\sigma = 0.8$). Figure EC.2 repeats the analysis at $\sigma = 0.8$. The CV procedure achieves over 94% of oracle revenue at all sample sizes, demonstrating robustness to heterogeneity.

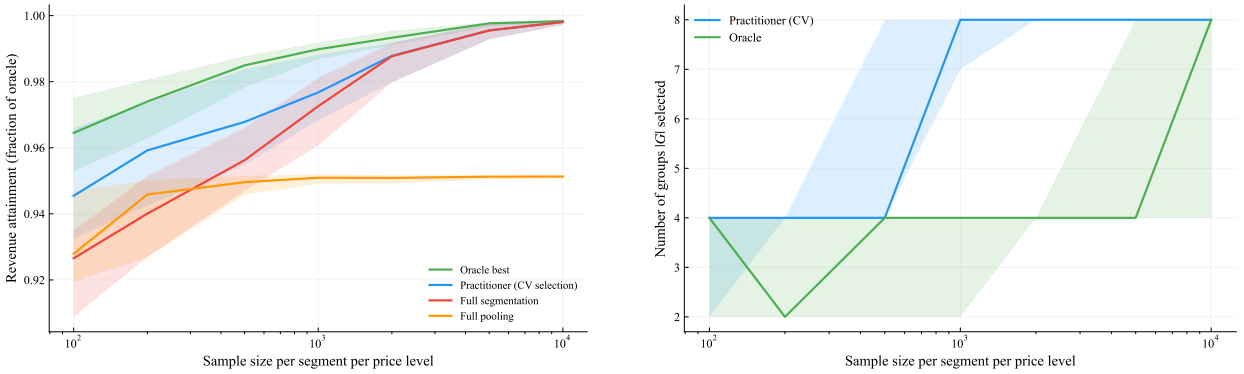


Figure EC.2 CV validation under high heterogeneity ($\sigma = 0.8$). *Left*: revenue attainment (CV reaches $\geq 94\%$ of oracle). *Right*: progressive refinement, where the oracle shifts toward finer segmentation as n falls and CV tracks the shift. Bands report IQR over 100 replications.

EC.8. Feature-Based vs. Segment-Based Pricing by Sample Size

To compare feature-based and segment-based pricing across ODs of varying size, we compute the per-OD difference in treatment effect and scale it by the average base price for cross-market comparability. We relate this scaled difference to log sample size (the per-OD plot is omitted from this public version). Above a threshold of $\log n \approx 10$ (about 22,000 observations), the two methods perform similarly, suggesting that with sufficient data both adequately capture demand and substitution; below the threshold, feature-based pricing

tends to outperform segment-based pricing, and the linear fit confirms a negative relationship between sample size and the performance gap.

EC.9. Heterogeneous-Cost Model versus MNL: Consistency and Misspecification Floor

This section reports the synthetic comparison summarized in §6 of the main text. The tier-elasticity gradient documented in §2 implies substantial variation in product-level substitution curvature. The experiment below quantifies the revenue cost of an MNL specification that imposes a common α across products, in a setting where the data-generating process and the optimal price are both known.

Setup. Data are generated from the heterogeneous-attention model with $I = 3$ products, $L = 4$ price levels, and a single customer segment. Controlled heterogeneity is introduced through $\alpha_i = \max(\alpha_{\text{base}} + \sigma_\alpha \cdot \delta_i,\ 0.2)$ with $\alpha_{\text{base}} = 1.2$ and a fixed perturbation $\boldsymbol{\delta}$; MNL is correctly specified at $\sigma_\alpha = 0$. For each $\sigma_\alpha \in [0, 1.5]$, both models are calibrated by LAD, and calibration loss, pricing error, and revenue loss are evaluated over 500 replications.

Noiseless calibration and parameter recovery. We first examine MNL's performance in the absence of estimation noise (true shares used directly) to isolate the misspecification effect. The left panel of Figure EC.3 plots MNL's LAD calibration residual versus σ_α. The residual is zero at $\sigma_\alpha = 0$ (where MNL is correctly specified) and grows monotonically (from 0 to 2.94) as α-heterogeneity increases, reflecting MNL's structural inability to fit product-specific α_i values. This residual is irreducible, and no amount of additional data can eliminate it. The right panel of Figure EC.3 visualizes the underlying mechanism: at each heterogeneity level, MNL's estimated $\hat{\alpha}$ (common across products) is compared against the true product-specific α_i values. The discrepancy between the flat MNL estimate and the spread of true values grows with σ_α.

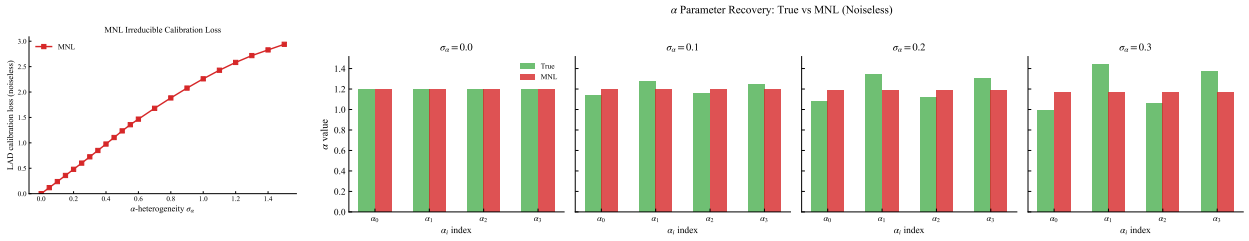


Figure EC.3 MNL noiseless calibration and parameter recovery. *Left*: MNL LAD residual vs. σ_α (the irreducible misspecification loss). *Right*: true α_i vs. MNL estimates at selected heterogeneity levels.

Pricing and revenue consequences. The parameter misspecification translates into pricing errors and revenue losses. The left panel of Figure EC.4 shows pricing error and revenue loss versus σ_α at a fixed sample size $n = 2{,}000$, with median and interquartile range over 500 replications. Our model's errors remain small across all heterogeneity levels because it can fit the true product-specific α_i. MNL's pricing error and revenue loss, by contrast, grow monotonically with σ_α: at $\sigma_\alpha = 0$ MNL has a small estimation

advantage (0.03% vs. 0.13% revenue loss), but at $\sigma_\alpha = 0.5$ MNL's revenue loss exceeds the heterogeneous-attention model's by roughly an order of magnitude (9×). The right panel of Figure EC.4 fixes $\sigma_\alpha = 1.0$ and varies the sample size. Both models' estimation errors decrease with n, but they converge to different limits: our model's errors approach zero (consistent estimation), while MNL's errors converge to a positive misspecification floor (dotted line) equal to its noiseless loss. This floor reflects the irreducible bias from the common-α restriction. Since α_i governs both substitution curvature and the product-specific markup in (5), even moderate misspecification propagates into systematic mispricing. The heterogeneous-attention model therefore incurs no penalty when α-heterogeneity is absent and delivers substantial gains when it is present.

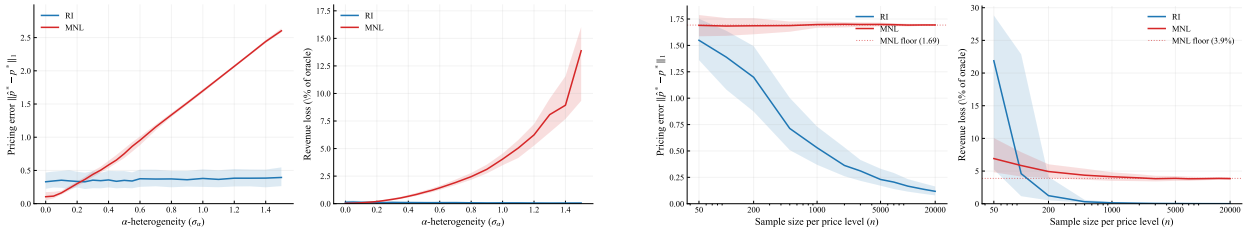


Figure EC.4 MNL misspecification: pricing error and revenue loss. *Left*: scaling with σ_α at $n = 2{,}000$ (median, IQR over 500 reps). *Right*: scaling with n at $\sigma_\alpha = 1.0$; MNL converges to a positive floor (dotted), our model to zero.

Irreducible floor, diagnostics, and revenue ratio. The left panel of Figure EC.5 plots MNL's best-case (noiseless, infinite-data) pricing error and revenue loss as a function of σ_α. These represent the minimum achievable loss under MNL regardless of sample size, and they grow with σ_α, consistent with the quadratic revenue loss bound of Theorem 5. The middle panel of Figure EC.5 shows that MNL's in-sample calibration loss at $n = 2{,}000$ also increases with σ_α, providing a practical diagnostic: a large in-sample residual signals that the common-α restriction is binding and that the heterogeneous-cost model may be warranted. Finally, the right panel of Figure EC.5 reports the revenue loss ratio (MNL / our model) as a function of σ_α. At low heterogeneity ($\sigma_\alpha \le 0.3$), both models perform similarly (ratio ≈ 1). At $\sigma_\alpha = 1.5$, the median ratio exceeds 200×, demonstrating that the practical cost of MNL misspecification can be substantial when product-level demand heterogeneity is high.

EC.10. Pilot Studies

Two pilot experiments preceded the main field deployment, both on a single short-haul route with booking-session-level randomization. Pilot 1 tested feature-based (FEA) pricing and the PCP rule against a uniform-pricing control (45%/10%/45% allocation). Pilot 2 tested FEA, segment-based (SEG), and PCP pricing against a multi-armed bandit (MAB) baseline (15%/15%/40%/30% allocation). Table EC.2 reports the estimated per-session revenue treatment effect under the full-control specification.

The pilots are small-sample, single-route exercises and no point estimate is significant at conventional levels; we report them only for the one pattern that bears on the theory. In Pilot 2, SEG is the sole arm with

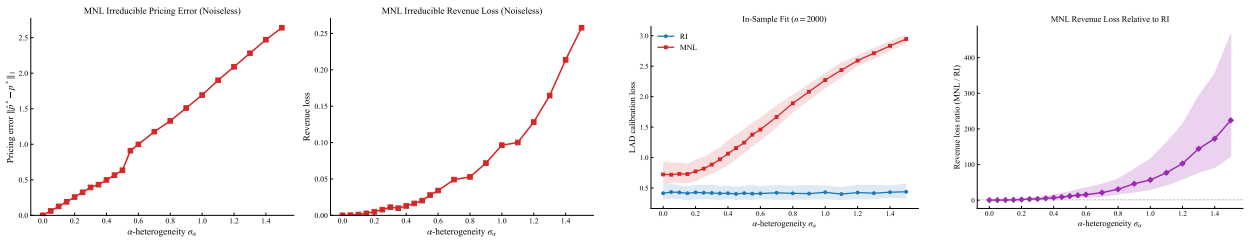


Figure EC.5 MNL irreducible floor, diagnostic, and revenue-loss ratio. *Left*: noiseless pricing error and revenue loss vs. σ_α; the floor grows with heterogeneity (cf. Theorem 5). *Middle*: in-sample calibration loss at $n = 2{,}000$ as a diagnostic. *Right*: revenue-loss ratio (MNL / our model); at $\sigma_\alpha = 1.5$ the median exceeds 200×.

Table EC.2 Estimated treatment effect on revenue in two pilots

	Pilot 1		Pilot 2		
	(1) FEA	(2) PCP	(3) FEA	(4) SEG	(5) PCP
Treatment	0.326	0.656	0.397	−1.635*	0.170
	(0.945)	(1.578)	(0.950)	(0.897)	(0.744)
Controls	Yes	Yes	Yes	Yes	Yes
Feature FE	Yes	Yes	Yes	Yes	Yes
OD FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes
Observations	2,706	1,707	3,634	3,657	4,656
PNRs	2,047	1,316	2,653	2,657	3,469
Adjusted R^2	0.172	0.184	0.245	0.244	0.206

Notes. Pilot 1 baseline is uniform pricing. Pilot 2 baseline is multi-armed bandit. Controls include the number of passengers, trip days, days to departure, airfare, as well as fixed effects of day-of-week, trip type, flight type, fare class, payment currency, and aircraft type; Date and Feature FEs as indicated. Cluster-adjusted standard errors at PNR level in parentheses. * $p < 0.10$

a (weakly significant) *negative* estimate, consistent with the data-scarcity regime of Theorem 3: with few observations per fine segment, estimation error dominates merging bias and full segmentation is penalized. FEA, which pools through a feature-linear shifter, avoids this penalty in both pilots. The main experiment's larger per-OD samples close this gap, so FEA and SEG perform comparably there (§6.2).

EC.11. Field-Test Implementation Details

Running the framework in the field test required addressing three implementation trade-offs beyond the paper's theoretical analysis. First, the unconstrained semi-closed-form solution lay within the Airline's $[-5\%, +20\%]$ price bounds for 60–70% of segment–product combinations; for the remaining cases, the field-test implementation invoked a four-stage validation chain (price-bound clamping, monotone ladder restoration, per-kg fairness projection, and a profitability guard that falls back to base prices). Second, the paper's calibration minimizes the KKT residual $\|\hat{A}_k\theta_k - \nu\|_1$, a price-discrepancy criterion; for the field test, we applied the Single Index (SI) refinement that alternates weighted least-squares recalibration of (α_i, β_i)

with LP refinement of feature parameters (up to 20 rounds, early-stopped when $\max|\Delta\alpha| + \max|\Delta\beta| < 10^{-6}$) to shift the objective to share-based predictive accuracy. Theorem 2 still applies: SI simply reduces the effective perturbation $\|E_k\|_{1\to 1}$ near the optimal shares. Third, the theoretical sample-size bound (Lemma 2), evaluated at $I = 5$, $\eta = 0.05$, $\epsilon = 0.5$, and a minimum observed share $s_{\min} \approx 0.4$, requires $n \geq 157$ per experiment, but the field test used $N_{\min} = 30$, which sufficed due to LAD robustness, SI stabilization, and the profitability guard; Laplace smoothing regularized the log-share singularity when an observed share was zero.

EC.12. Customer Strategic Behavior

Treatment noncompliance arises when customers exploit price variation by browsing multiple sessions. Cookie-based randomization mitigates this risk, but post-booking sessions remain susceptible.

To assess such behavior, we analyze customers' browsing patterns in post-booking sessions, where the same booking can be revisited and different prices observed. We compare sessions that result in a purchase to those that do not.

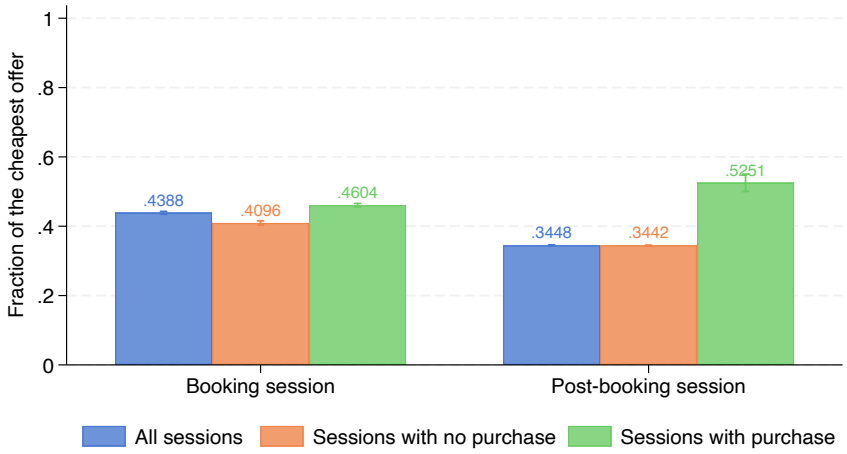


Figure EC.6 Fraction of sessions showing the cheapest offer, split by purchase decision.

In Figure EC.6, we examine whether customers purchase when seeing the cheapest offer by depicting the fraction of sessions with the cheapest offer (offering base product with the lowest price) the customer ever saw. In post-booking sessions, we observe a much higher fraction in sessions with purchase compared to the ones without purchase (52.51% vs. 34.42%), indicating strategic behavior, while the fraction for sessions without purchase is highly consistent with all sessions (34.48%). For booking sessions, since we do not observe browsing history, we proxy the cheapest offer of a customer by the offer with lowest price for the corresponding customer segment (identified by feature combinations). Since this is an imperfect proxy, we

observe some variation in the cheapest offer fraction across session types, but the fraction is close for all sessions and sessions with or without purchase (43.88% vs. 40.96% vs. 46.04%).

These results suggest that a considerable proportion of post-booking purchases are subject to strategic purchase behavior. The estimates of treatment effect in post-booking sessions are likely driven by price shopping and thus biased in favor of the cheaper treatment. Booking sessions are less susceptible to such bias.

EC.13. Heterogeneous Treatment Effects

The field-experiment treatment effects vary across booking populations along two primary dimensions: passenger region and flight-haul length. The effect of feature-based pricing relative to control is positive in all regions except two, both with the smallest sample sizes. Flight duration is a primary dimension of market heterogeneity: shorter ODs have lower baggage demand and higher price sensitivity, consistent with the positive trip-days and airfare coefficients in Table 3. Classifying ODs as short-, medium- or long-haul by the Airline's haul definition, both methods significantly outperform the control group in medium-haul ODs, with lower performance in short-haul ones. In long-haul ODs, both methods yield positive but insignificant effects, likely due to the smaller sample sizes. The feature-based method has a lower point estimate than the segment-based method on these routes, consistent with feature pooling muting the route-specific patterns that segment-level estimation captures. Per-region, per-OD and per-haul plots are omitted from this public version.

EC.14. Managerial Insights from Calibrated Parameters

This section details the auxiliary parameter-fitting exercise summarized in §6.4: a historical extract of directional ODs, separate from the field-test data, of which 30 stabilized within the convergence threshold. The patterns below report structural managerial insights from the calibrated parameters.

Distance gradient in attention costs. Calibrated α_i rises monotonically with flight time (Table EC.3), from a median of ≈ 24 on very short-haul routes to ≈ 170 on long-haul routes, a sevenfold spread. Since α_i enters the optimal price (5) as a product-specific markup, this maps directly into pricing posture: short-haul routes support tight, competitive ladders, long-haul routes sustain premium markups. Within each distance tier the α profile is approximately flat across the 20–40 kg tiers, supporting the reading of α as a route-context primitive rather than a tier-specific one. Return-leg routes show systematically higher median α (≈ 100) than outbound routes (≈ 55), consistent with committed-luggage demand on the return leg.

Tier differentiation and feature levers. Two further patterns are actionable. First, intrinsic-value spreads $v_{40\,\mathrm{kg}} - v_{20\,\mathrm{kg}}$ widen from ≈ 50 utility units on very short-haul routes to ≈ 250 on long-haul routes; with Theorem 1, this licenses aggressive tier differentiation on long routes and a compressed ladder on short ones. Second, feature-utility analysis identifies the levers for the feature-linear shifter: *days-to-departure* (+1.2 to

Table EC.3 Median calibrated α_i by route distance tier (well-calibrated markets, $n = 30$).

Route tier	**Median α**	**Interpretation**
Very short-haul	23.6	Highly price-sensitive
Short haul	32.7	Price-sensitive
Medium haul	89.1	Moderately sensitive
Long haul	170.3	Least sensitive

+4.2, strongest long-haul) and *booking class* (+1.0 to +4.0) are uniformly positive, enabling timing- and fare-class-based discrimination; *seasonal effects* concentrate on long-haul routes (+1.9 to +3.7); *connecting passenger* is uniformly positive (+0.8 to +1.9); and *one-way trip* reverses sign by route. All cross-route comparisons use $\gamma_k = 1$ and assume approximately homogeneous price sensitivity across routes.